

STDA - A3 - Spatio-Temporal Analysis of Global Seismicity (2000 - 2025)

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Abstract—This study presents a comprehensive spatio-temporal analysis of global seismic activity from 2000 to 2025, integrating both spatial and temporal perspectives to better understand earthquake behavior. Spatial analysis reveals strong clustering along tectonic boundaries, confirming significant spatial dependence and localized seismic hazards. Temporal modeling highlights the clustered nature of earthquakes, with clear aftershock patterns and regime shifts captured through advanced machine learning techniques. By combining these dimensions, the study demonstrates that seismic activity is driven by both geographic structure and temporal dynamics rather than independent random processes. Density-based clustering and self-exciting models effectively capture real-world seismic patterns, while predictive models highlight the limitations of using static features alone. The findings reinforce the importance of integrated feature engineering for capturing complex natural phenomena. Overall, this work provides a structured framework for analyzing seismic data and offers insights for improved hazard assessment and future predictive modeling.

Index Terms—Spatio-Temporal Analysis, Seismic Activity, Machine Learning, Spatial Clustering

I. SPATIAL TASKS

The study of earthquakes presents a canonical use case for spatial data mining. Unlike standard tabular data where observations are assumed to be independent, geographic data exhibits spatial dependence and heterogeneity. Guided by Tobler's First Law of Geography [2], which states that near things are more related than distant things, this section explores the profound spatial autocorrelation, clustering tendencies, and colocation patterns inherent in tectonic activities. The analysis strictly adheres to the Spatial Data Mining (SDM) Pyramid, progressing from raw spatial data to spatial information through map-matching and synthesis, and finally to spatial knowledge via rigorous statistical testing and machine learning architectures.

A. Task 1: Spatial Feature Engineering

Spatial feature engineering involves transforming implicit spatial relationships into explicit variables that algorithms can process.

1) *Global Distribution and Exploratory Attributes*: To establish a baseline, we first visualize the raw spatial distribution of the dataset. **Figure 1** illustrates the global map of earthquakes. The stark, non-random alignment of seismic events immediately highlights the underlying tectonic plate

boundaries, particularly the Pacific Ring of Fire and the Mid-Atlantic Ridge. **Figure 2** shows right-skewed distributions for magnitude and depth, indicating that low-magnitude, shallow earthquakes are far more frequent, consistent with established seismic occurrence patterns such as the Gutenberg-Richter relationship [3].

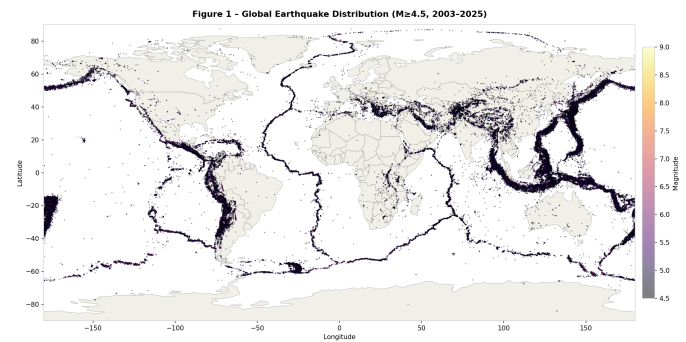


Fig. 1. Global spatial distribution of 175,946 seismic events.

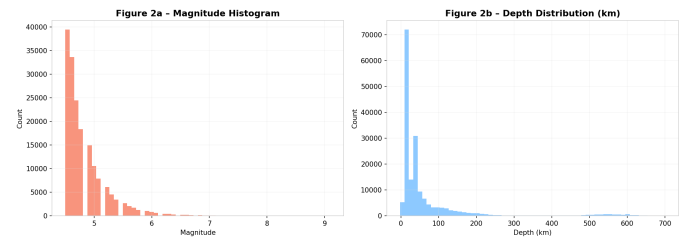


Fig. 2. Histograms of magnitude and depth.

2) *Depth Categorization and Tectonic Zones*: Depth is a critical spatial feature. We engineered a categorical feature mapping depths into *Shallow* (< 70 km), *Intermediate* (70 - 300 km), and *Deep* (> 300 km). **Figure 3** maps these categories, revealing that deep earthquakes are exclusively clustered around subduction zones. Furthermore, geographic coordinates were binned into localized *Tectonic Zones*. **Figure 4** visualizes these engineered boundaries, transitioning the data from a continuous coordinate space to a discrete, regional feature space.

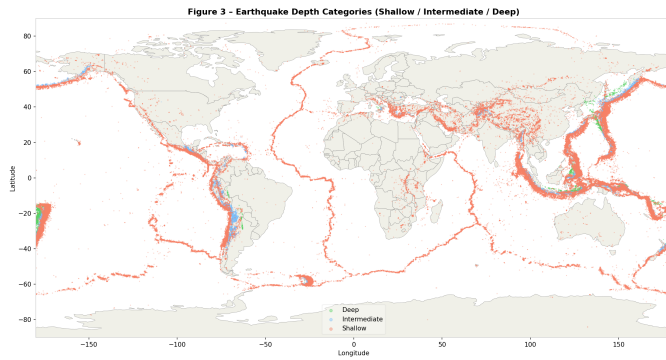


Fig. 3. Earthquake distribution colored by depth categories.

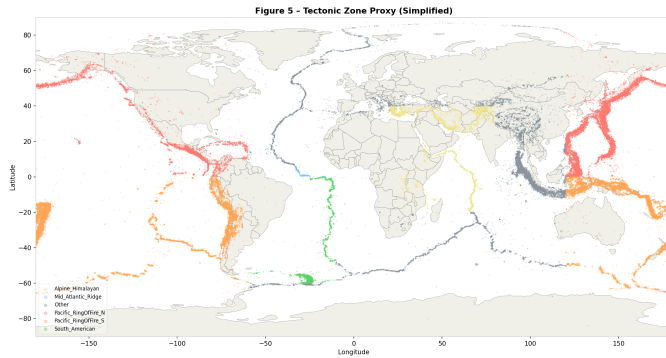


Fig. 4. Discrete spatial binning of events into localized tectonic zones.

3) *Regional Aggregations and Proximity Metrics*: To incorporate neighborhood effects, we calculated the *Regional Magnitude Mean* using grid-based aggregations. As shown in Figure 5, the regional mean magnitude appears relatively uniform across grid cells, indicating limited spatial variation despite clear clustering of seismic activity along tectonic boundaries.

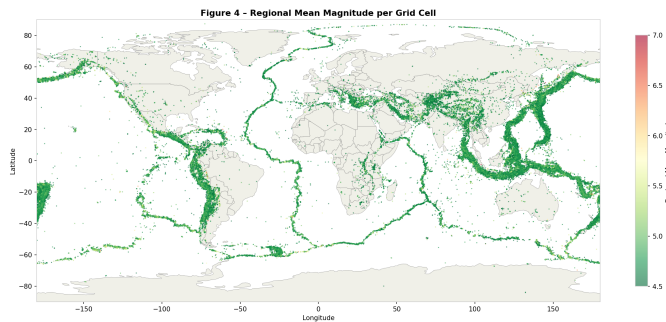


Fig. 5. Spatial distribution of the regional magnitude mean feature.

Distance-based metrics are fundamental to spatial analysis. **Figure 6** displays the distribution of Nearest Neighbor (NN) distances computed via a KD-Tree. The heavy concentration of events with near-zero NN distances suggests extreme spatial aggregation, violating independent distribution assumptions.

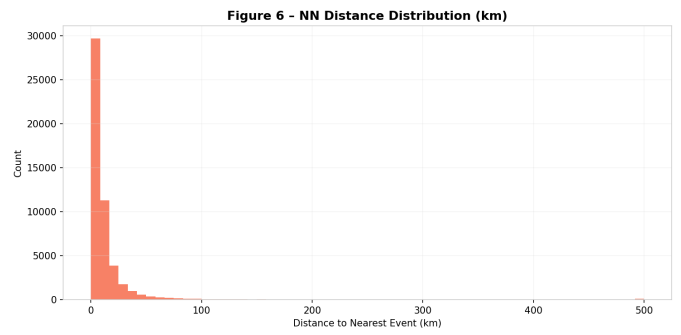


Fig. 6. Distribution of spatial Nearest Neighbor (NN) distances.

4) *Magnitude and Depth Correlations*: We further explored the relationship between our engineered features. Figure 7 shows that the largest magnitude earthquakes are predominantly shallow, while deep earthquakes are more tightly clustered around moderate magnitudes with less variability. Figure 8 confirms the Gutenberg-Richter relationship, showing that earthquake frequency decreases exponentially with increasing magnitude. Figures 9 and 10 highlight variations in depth distribution across tectonic zones, with subduction regions, particularly along the southern Pacific Ring of Fire, showing a higher proportion of deep and intermediate earthquakes.

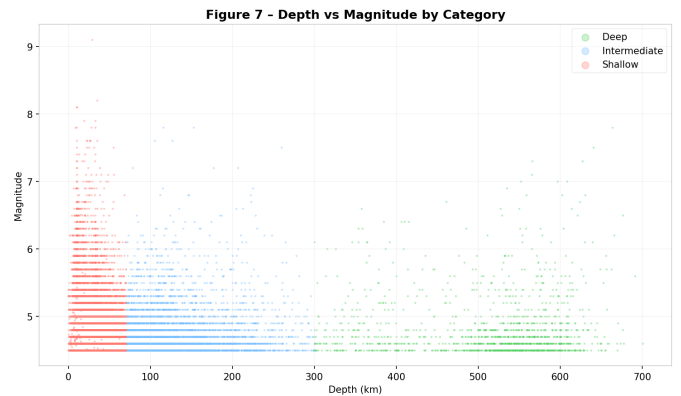


Fig. 7. Scatterplot exploring the relationship between depth and magnitude.

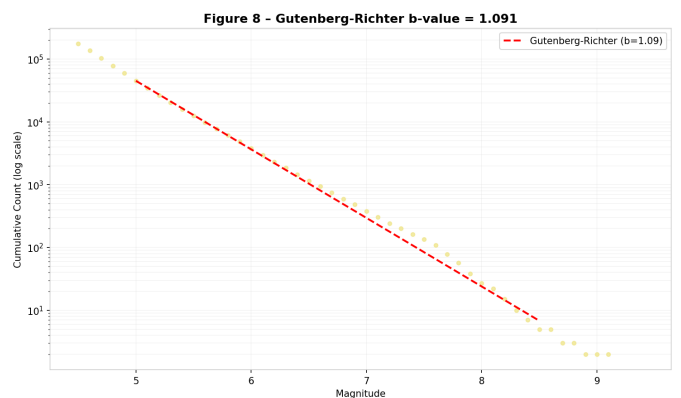


Fig. 8. Gutenberg-Richter magnitude-frequency distribution.

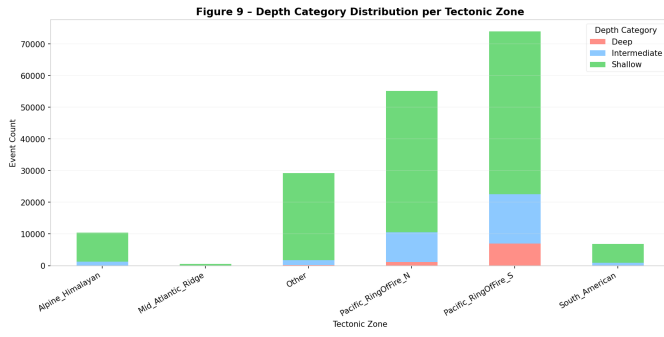


Fig. 9. Distribution of earthquake depths across varying tectonic zones.

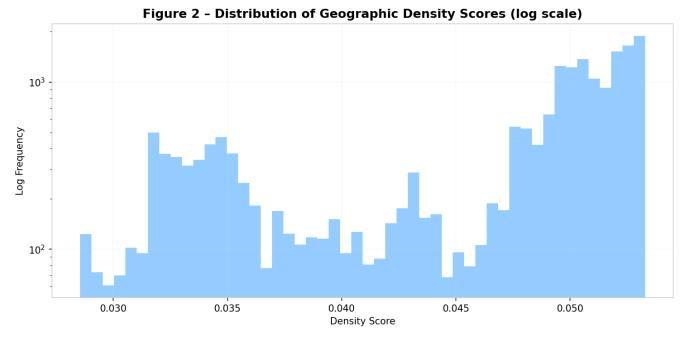


Fig. 12. Histogram of spatial density values derived from the KDE model.

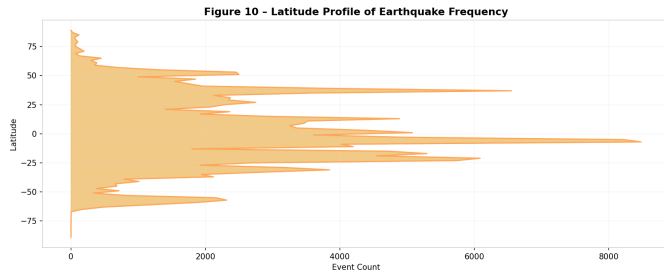


Fig. 10. Latitudinal profile mapping event frequency and average depth.

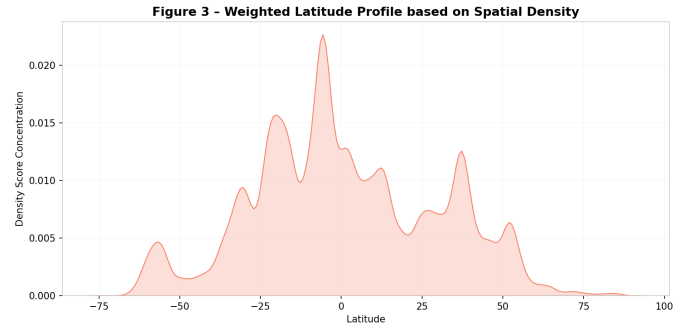


Fig. 13. Latitudinal profile of KDE spatial intensity.

B. Task 1: Spatial Machine Learning Models

With a robust set of engineered spatial features (spatial lag, tectonic zone, NN distance), we deployed machine learning models to identify spatial patterns and predict seismic properties.

1) *Spatial Density Estimation*: We began with Kernel Density Estimation (KDE) to generate a continuous probability surface of seismic occurrences. **Figure 11** visualizes the KDE map, effectively translating point-process data into an intensity field. **Figure 12** and **Figure 13** further quantify this intensity, demonstrating that spatial density is not normally distributed but rather characterized by extreme peaks along specific latitudinal fault lines.

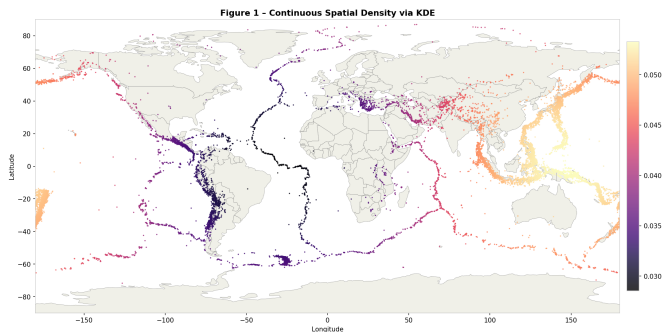


Fig. 11. Kernel Density Estimation (KDE) map of global seismicity.

2) *Density-Based and Probabilistic Clustering*: Traditional algorithms like K-Means struggle with spatial data due to its non-convex, linear nature (fault lines). Therefore, we applied Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). Clustering was performed on 3D spatial coordinates (Latitude, Longitude, and Depth) to accurately capture subduction zones. As shown in **Figure 14**, HDBSCAN successfully isolated 165 dynamic-density clusters representing major fault systems, while designating background seismicity as noise. **Figure 15** visualizes the cluster membership probabilities. **Figure 16** highlights HDBSCAN's superiority over standard DBSCAN. While standard DBSCAN merges almost all events into a single massive cluster due to varying global densities, HDBSCAN successfully segments the data into distinctly sized tectonic groups.

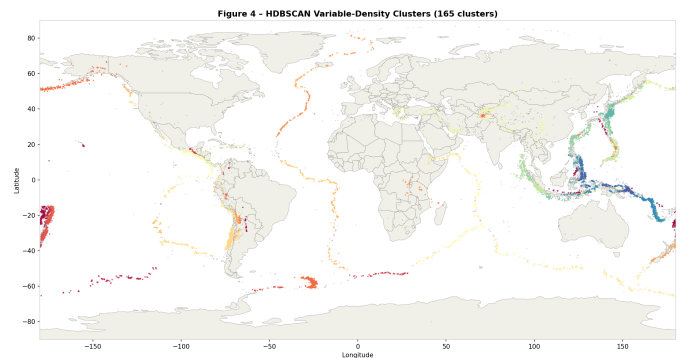


Fig. 14. HDBSCAN cluster assignments mapping distinct fault line structures.

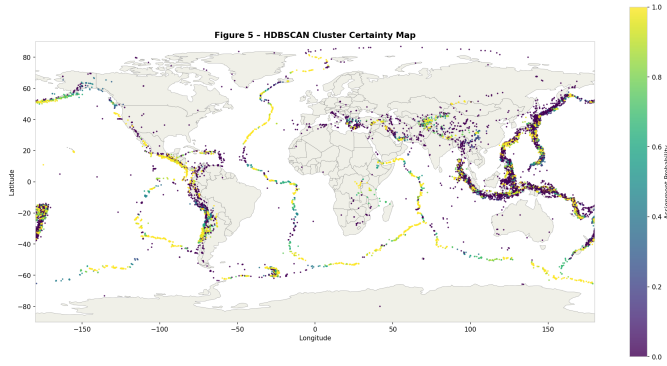


Fig. 15. HDBSCAN membership probabilities highlighting transition zones.

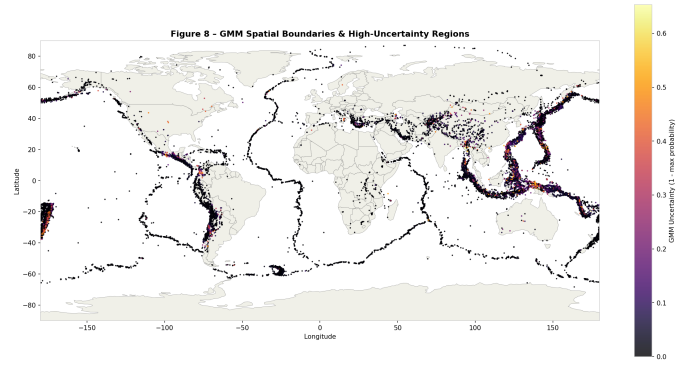


Fig. 18. Spatial uncertainty of GMM assignments across global geography.

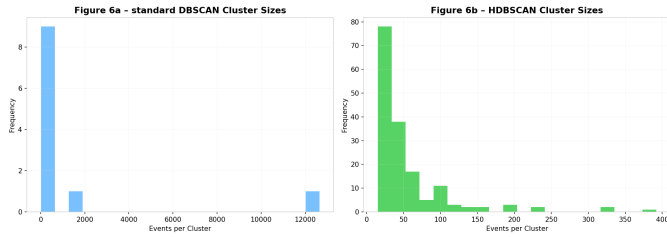


Fig. 16. Comparison of cluster sizes determined by HDBSCAN.

For a soft-clustering alternative, a Gaussian Mixture Model (GMM) with 15 components ($K=15$) was fit. **Figure 17** displays the GMM spatial clusters. Unlike HDBSCAN, GMM forces all points into probability distributions. **Figure 18** plots the uncertainty of the GMM assignments, which is highest precisely at the boundaries of colliding tectonic plates, proving the model's sensitivity to spatial transitions.

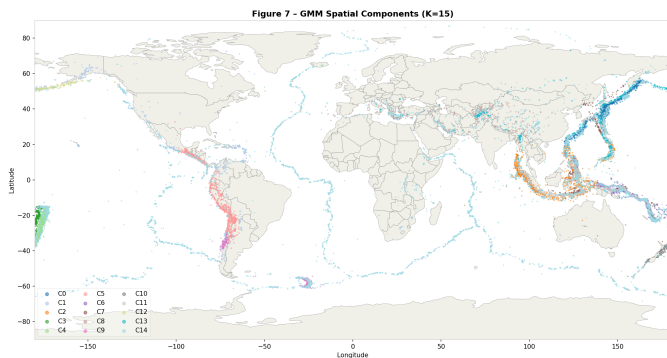


Fig. 17. Spatial domains identified through Gaussian Mixture Modeling (GMM).

3) *Predictive Spatial Modeling (XGBoost & Random Forest)*: We utilized XGBoost and Random Forest to predict earthquake magnitudes using *only* spatial predictors. While absolute prediction is notoriously difficult, feature importance analysis is highly revealing. **Figure 19** shows that the event's depth, followed by absolute geographic coordinates (latitude/longitude), are the most critical features for XGBoost, indicating that the 3D location is the strongest determinant of baseline magnitude. **Figure 20** confirms that residuals are roughly centered, though a long right tail exists, indicating that the model systematically under-predicts the highest magnitude (extreme) events.

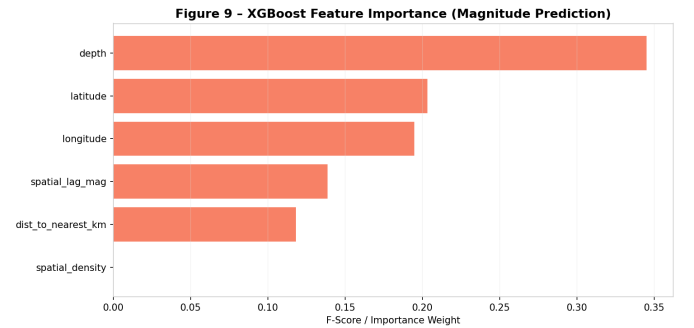


Fig. 19. Feature importance plot from the XGBoost spatial predictive model.

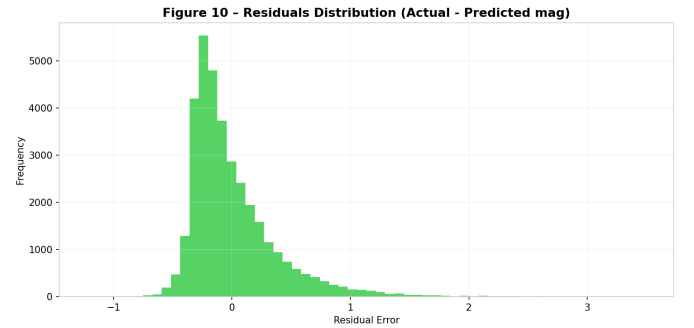


Fig. 20. Residual analysis of the XGBoost predictions.

The Random Forest model demonstrated slightly better non-linear mapping capabilities. **Figure 21** visualizes the Random

Forest’s non-linear response to depth changes at a fixed coordinate (the Japan subduction zone). The step-like jumps in predicted magnitude perfectly illustrate the tree-based model capturing distinct tectonic depth thresholds.

Figure 22 compares the Mean Squared Error (MSE) and R^2 scores of both models. While the Random Forest model marginally outperforms XGBoost, the overall R^2 scores remain objectively low (0.057 and 0.076). Rather than a failure of the algorithm, this mathematically illustrates a known seismological reality: static spatial coordinates alone are insufficient to predict precise earthquake magnitudes. Magnitude is primarily driven by chaotic, time-dependent stress accumulations and fault-slip mechanics. Therefore, these models successfully capture the underlying spatial “baseline” signal, proving that certain tectonic regions have different magnitude carrying capacities, but confirm that robust magnitude prediction inherently requires temporal, sequence-based, or physical sensor data.

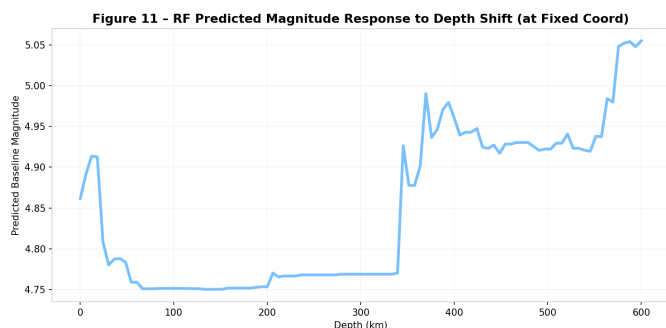


Fig. 21. Random Forest response mapping the non-linear impact of depth.

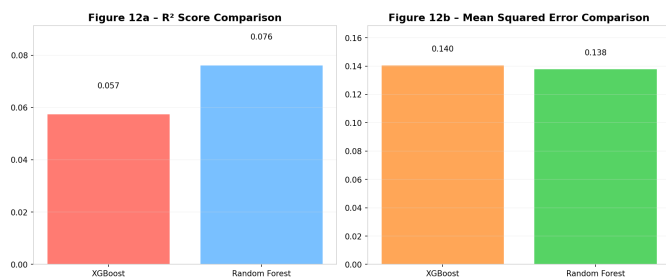


Fig. 22. Performance comparison between XGBoost and Random Forest models.

C. Task 3: Spatial Data Mining

Task 3 uses specialized spatial algorithms to uncover clustering patterns, spatial autocorrelation, colocation and hotspots, contrasting them against baseline assumptions.

1) *Baseline Clustering and 3D Extensions:* To emphasize the uniqueness of spatial data, we first evaluated the optimal number of clusters using the Elbow Method as shown in **Figure 23**, identifying a clear inflection point at $k = 7$. We then ran K-Means utilizing these 7 clusters in **Figure 24**. However, K-Means imposes artificial, spherical Voronoi tessellations that ignore the continuous, linear nature of actual

fault systems. Similarly, a baseline Gaussian Mixture Model utilizing 8 components as depicted by **Figure 25** creates large, generic regional blobs. To effectively capture spatial reality, we executed a 3D DBSCAN utilizing Latitude, Longitude, and Depth. **Figure 26** highlights these 3D clusters, effectively separating shallow crustal quakes from deep subduction phenomena in the same 2D coordinate space.

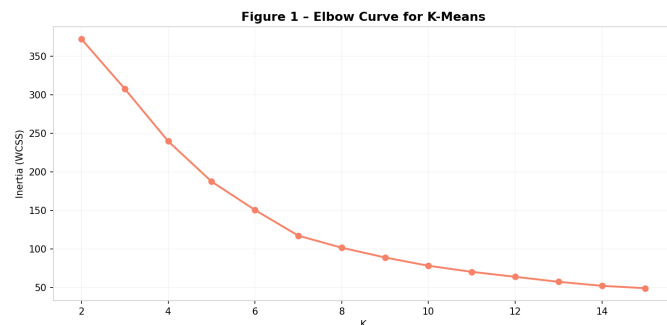


Fig. 23. Elbow plot used for determining baseline K-Means optimal clusters at $k = 7$.

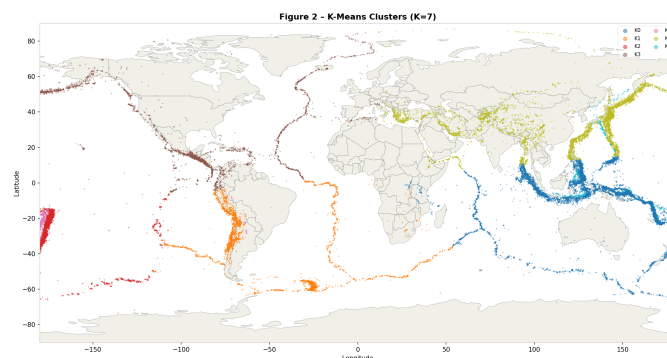


Fig. 24. Baseline K-Means clustering ($k = 7$) demonstrating artificial spatial partitioning.

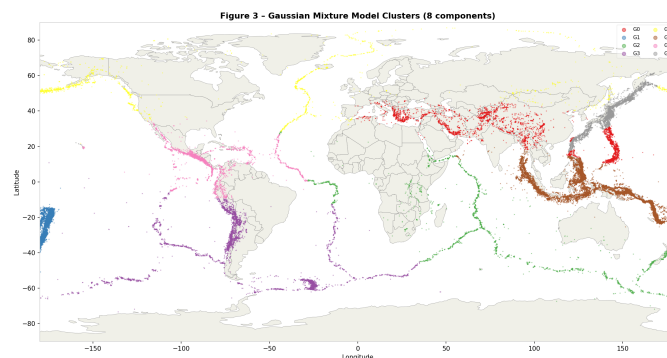


Fig. 25. Baseline GMM spatial clustering lacking fault-line granularity.

Figure 16 - 3-D DBSCAN Clusters (top 6)

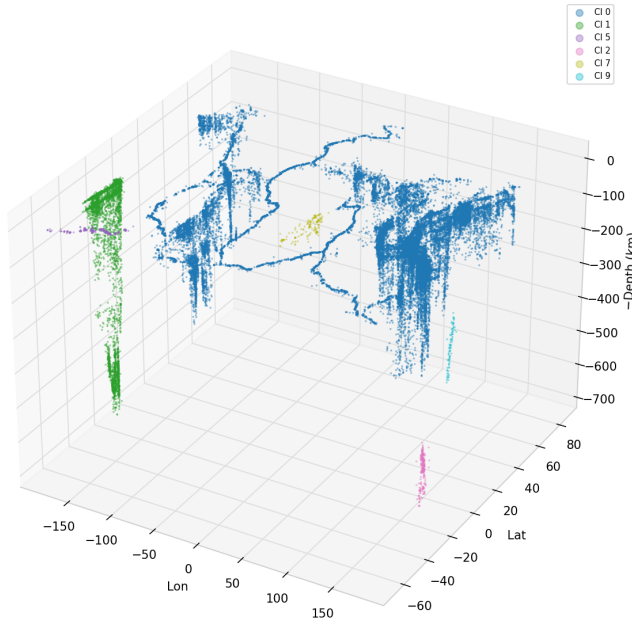


Fig. 26. 3D DBSCAN clustering utilizing latitude, longitude, and depth simultaneously.

2) *Spatial Outliers and Autocorrelation*: A spatial outlier differs significantly from its physical neighbors. We calculated the spatial Z-score for magnitude. Events with $|Z| > 2.0$ were flagged as spatial outliers (2,996 events). **Figure 27** plots these anomalies: unexpectedly large earthquakes in quiet zones, or notably small earthquakes in active zones.

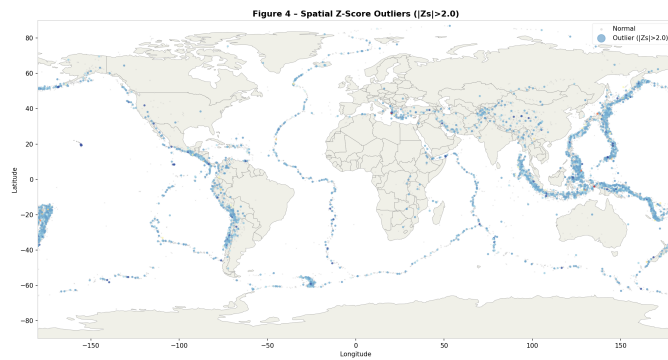


Fig. 27. Spatial outliers identified via localized Z-scores ($|Z| > 2.0$).

To quantify Tobler's First Law, we computed Global Moran's I. **Figure 28** shows the Moran scatterplot yielding $I = 0.044$. While the magnitude is small, it indicates positive spatial autocorrelation, high magnitudes slightly cluster with high magnitudes. **Figure 29** displays the empirical variogram, tracking variance as a function of spatial distance, proving that spatial dependence decays predictably over range.

Figure 5 - Moran Scatterplot of Magnitude

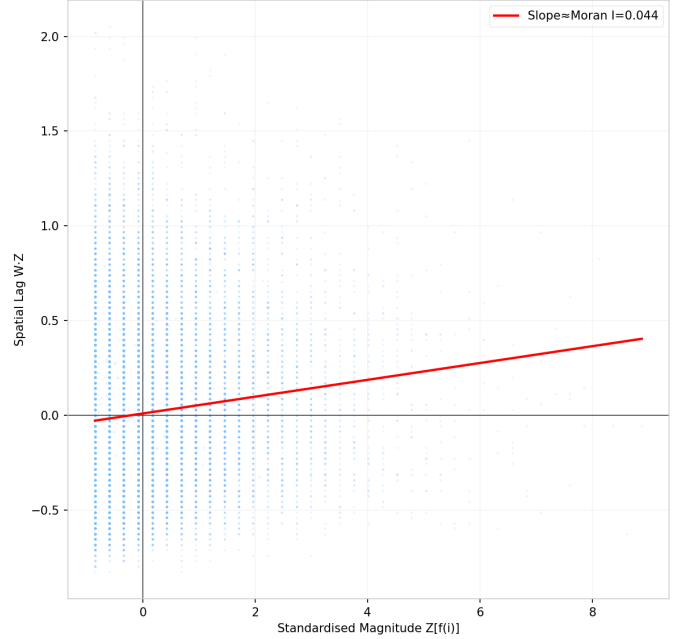


Fig. 28. Moran's scatterplot evaluating global spatial autocorrelation.

Figure 6 - Variogram Cloud (Magnitude)

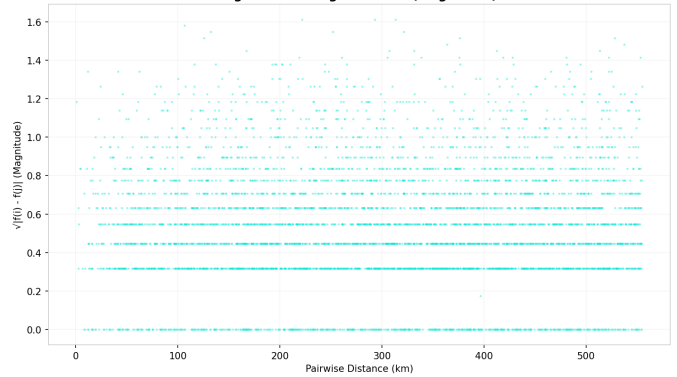


Fig. 29. Empirical variogram modeling spatial dependence decay over distance.

3) *Spatial Colocation Mining*: Colocation mining identifies boolean spatial features that frequently co-occur in the same neighborhood. We defined boolean features: *High Magnitude* (> 6.0), *Mid Magnitude*, *Shallow*, and *Very Deep*. **Figure 30** shows their individual prevalence.

Figure 7 - Boolean Spatial Features for Collocation

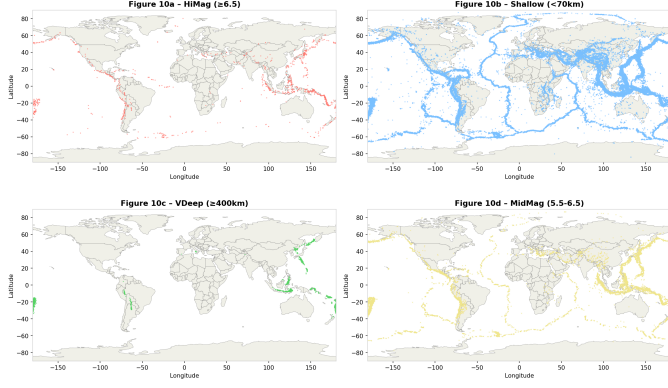


Fig. 30. Prevalence distributions of constructed boolean spatial features.

We utilized two distinct methodologies to mine collocation: grid-based association rules and event-centric neighborhood ratios. **Figure 31** maps the grid-level Support and Confidence logic, showing a high probability ($P=0.895$) of shallow events occurring in the same grid as high-magnitude events. We validated this at the micro-level using an event-centric participation index in **Figure 32**. Remarkably, looking within a tight 1.0° radius of any High Magnitude event, there is a 0.958 conditional probability of finding a Shallow event. In contrast, Very Deep events rarely collocate with High Magnitude events. **Figure 33** maps these precise collocation pairs, visually confirming the statistical findings along the Pacific Rim.

Figure 15 - Spatial Association Rule Matrix

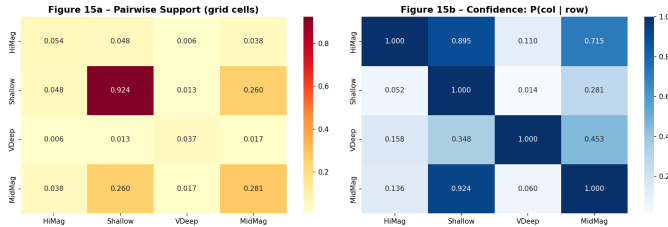


Fig. 31. Support and confidence metrics for spatial association rules.

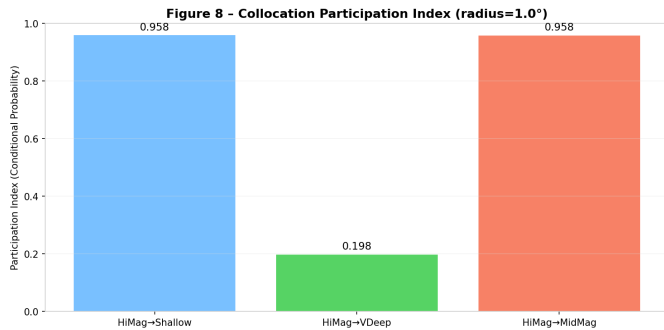


Fig. 32. Conditional probabilities demonstrating strong collocation between High Magnitude and Shallow depth.

Figure 9 - Co-location Map: High Magnitude & Shallow Events

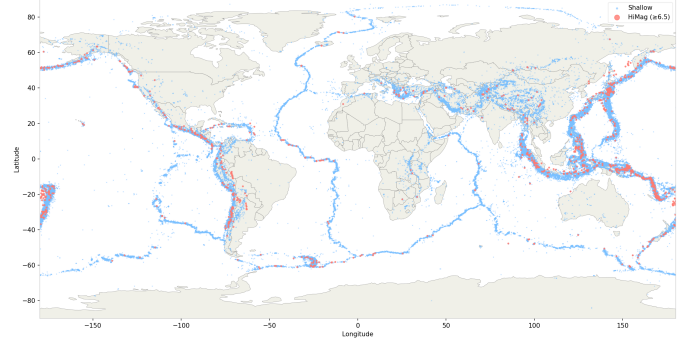


Fig. 33. Geographic mapping of collocated spatial features discovered via mining algorithms.

4) *Point Pattern Analysis and CSR*: To robustly test for clustering against Complete Spatial Randomness (CSR), we employed Ripley's K-function and its variance-stabilized transformation, Ripley's L-function. As seen in **Figure 34**, the observed $K(t)$ dramatically exceeds the theoretical CSR envelope across all distances. Furthermore, the L-function curve demonstrates that this clustering is highly statistically significant, confirming intense global spatial aggregation. We also explored a Cross-K function as seen in **Figure 35** to analyze distance-based relationships between different magnitude groups.

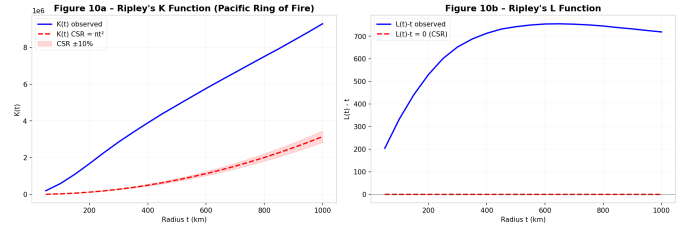


Fig. 34. Ripley's K-function analysis confirming global spatial aggregation against CSR.

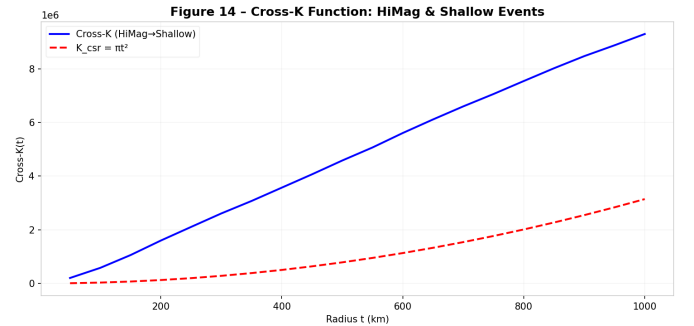


Fig. 35. Cross-K function analyzing inter-group spatial clustering tendencies.

Additionally, **Figure 36** depicting Nearest Neighbor Analysis (NNA) yielded an R-ratio of 0.331 ($R < 1$), further cementing the non-random, clustered nature of the data .

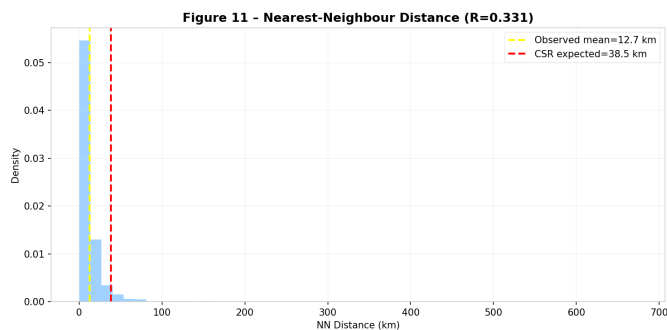


Fig. 36. Nearest Neighbor Analysis (NNA) results proving non-random aggregation.

5) *Hotspot Identification*: Finally, to identify significant localized clusters of high magnitude, we computed the Getis-Ord G_i^* statistic. **Figure 37** reveals 704 intense spatial hotspots. Unlike KDE, G_i^* considers the neighborhood statistical significance, pinpointing zones with highly hazardous seismic potential. **Figure 38** (Spatial Correlogram) provides the final piece of evidence, showing how local spatial correlation decays rapidly past 100km, validating the localized nature of the identified hotspots.

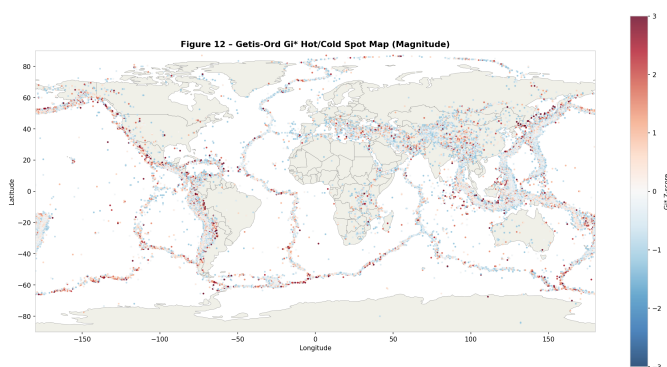


Fig. 37. Hotspot analysis via Getis-Ord G_i^* revealing statistically significant clusters.

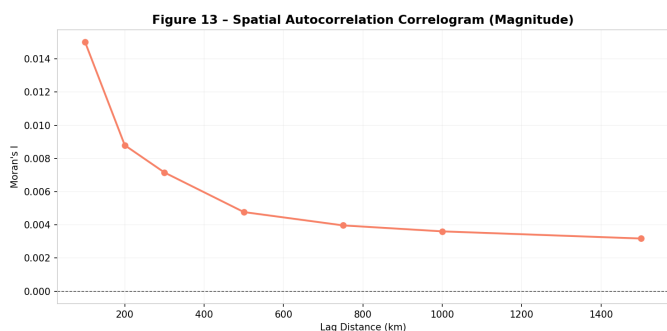


Fig. 38. Spatial correlogram mapping the decay of autocorrelation.

D. Analysis of Spatial Tasks

Looking back across the spatial tasks, this project really highlighted why geographic data requires specialized analytical tools. Standard datasets often assume that observations

are independent, but our early exploratory maps made it clear that earthquakes are tightly bound to tectonic structures. Our statistical tests in Task 3, specifically the Nearest Neighbor Analysis ($R=0.331$) and Ripley's K-function backed this up by completely rejecting Complete Spatial Randomness (CSR). Because the data is so heavily aggregated, standard algorithms like K-Means and GMM struggled, essentially drawing artificial borders that ignored the actual geology. Instead, we found that density-based approaches like HDBSCAN and 3D DBSCAN were much better suited for the job, as they could naturally trace the continuous, irregular fault lines.

The most interesting takeaway from this study was seeing the relationship between an earthquake's depth and its magnitude confirmed across different methodologies. In Task 1, our initial scatterplots and tectonic profiles hinted that the most devastating events occur close to the crust. In Task 3, we were able to actually quantify this using spatial colocation rules, finding a massive 95.8% conditional probability that a high-magnitude event will be shallow. Finally, when we built our predictive models, XGBoost independently confirmed this relationship, ranking 3D depth as the most important spatial feature for determining a region's baseline magnitude.

Ultimately, this analysis proved Tobler's First Law of Geography in a very literal sense: seismic hazard is highly localized. While our predictive models had low R^2 scores, this highlighted an important real-world limitation: static spatial coordinates are excellent for mapping out regional hazards, but predicting exact magnitudes requires time-series or physical sensor data. Even so, by engineering spatial features and applying localized point-pattern algorithms, we were able to successfully uncover the complex, structural patterns driving global seismic activity.

Future Spatial Analysis: While these tasks successfully extracted complex spatial patterns, exploring additional geographic dimensions could further improve our understanding of the dataset. For example, future work could integrate explicit geospatial vector data such as known fault-line geometries, tectonic plate boundaries (convergent, divergent, transform), and soil composition maps as physical covariates in the machine learning models. Additionally, migrating from grid-based density aggregations to 3D voxel-based spatial mining could provide a more precise volumetric analysis of deep subduction zones. Finally, employing spatial interpolation techniques, such as Kriging, or spatial graph-network analysis could help model how physical stress transfers across geographic distances between interconnected tectonic regions.

II. TEMPORAL TASKS

Earthquakes are inherently temporal phenomena, with foreshocks, mainshocks, and aftershocks unfolding over time. This section focuses on analyzing the temporal aspect of our global seismicity dataset. We define a robust feature engineering pipeline, employ suitable machine learning models to capture regime shifts and predict future trends, and utilize data mining tasks such as sequence mining and change point detection to extract underlying temporal patterns.

A. Task - 2 Temporal Feature Engineering

To adequately represent the temporal dynamics of seismicity, the raw timestamp data must be transformed into meaningful features that machine learning models can interpret.

1) *Basic Temporal Aggregation and Overview:* We began by analyzing the simple distribution of earthquakes over time to set a baseline for our modeling. As shown in Figure 39, the annual earthquake count reveals a significant spike in recorded seismic activity between 2010 and 2015. Additionally, the magnitude distribution shows that the vast majority of events cluster around the mean magnitude of 4.81, with frequency dropping off exponentially for larger earthquakes. The focal depth analysis further confirms that most of these events are relatively shallow, exhibiting a median depth of 30 kilometers.

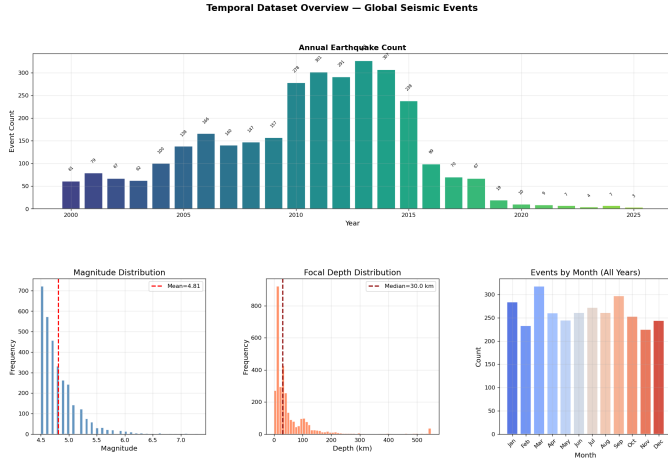


Fig. 39. Overview of the earthquake dataset, showcasing the raw event distribution and preliminary aggregated statistics.

2) *Cyclical Features:* Time intrinsically involves repeating cycles like days and months. Standard numerical encoding treats the transition from December to January as a large jump rather than a continuous loop. To fix this, we applied sine and cosine transformations to the temporal components. Figure 40 visualizes this approach perfectly. By mapping the time data onto polar coordinates, the scatter plots form closed circles. This mathematical adjustment ensures that our models understand that late night hours smoothly transition into early morning hours, preserving the true continuous nature of time.

3) *Aftershock Timelines and Inter-Event Times:* Earthquakes frequently occur in clusters, so measuring the elapsed time since the previous significant event helps us understand stress accumulation and release. Figure 41 clearly illustrates this clustering effect. In the top timeline, we can see dense clusters of aftershocks immediately following major magnitude 6 or greater events. The bottom distribution chart supports this by showing a steep drop in probability density as the days since the last major event increase. In short, once a large earthquake occurs, the probability of subsequent shocks is extremely high in the immediate following days.

4) *Rolling Statistics and Historical Context:* To provide our machine learning models with a sense of short-term memory,

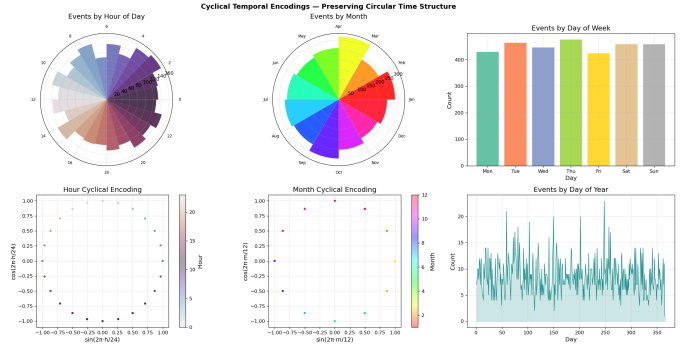


Fig. 40. Cyclical encodings (sine and cosine transformations) of the day of year, maintaining temporal continuity across boundaries.

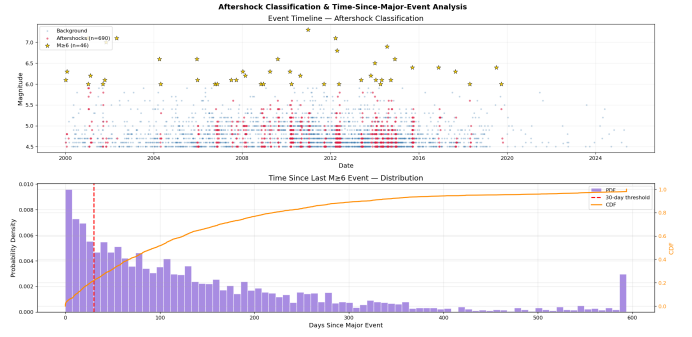


Fig. 41. Timeline of major aftershocks following significant seismic events, demonstrating the clustered nature of seismicity.

we engineered rolling statistical features. Figure 42 visualizes these moving averages over the entire timeline. The top plot displays the 7-day and 30-day rolling event counts. This smoothing technique clearly highlights the sustained period of intense global seismicity between 2010 and 2015, which is harder to see in raw daily counts. The bottom plot tracks the daily mean magnitude, with a reference line marking significant magnitude 6 events. By feeding these rolling windows into our models, we ensure the algorithms understand the immediate historical context of any given day rather than analyzing each day in total isolation.

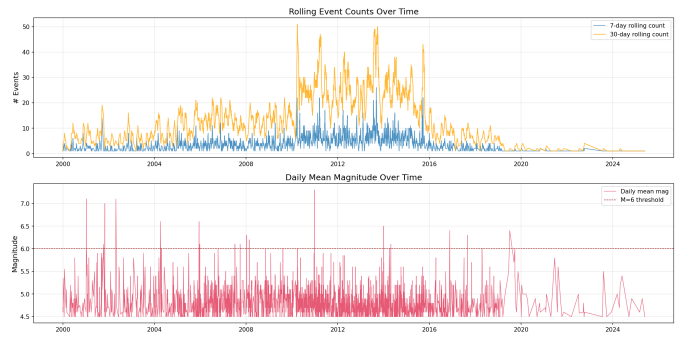


Fig. 42. Rolling event counts and daily mean magnitude, engineered to provide machine learning models with short-term historical context.

5) *Feature Distributions and Statistical Health:* Before feeding data into predictive models, it is crucial to understand the mathematical shape of our engineered features. Figure 43 displays these distributions. A key observation is the raw time between events, which is heavily skewed to the right. This means most earthquakes happen in rapid succession during clusters, while a few have long periods of absolute silence. By applying a logarithmic transformation to these intervals, we successfully created a more balanced, bell shaped distribution. This normalization step is vital because machine learning algorithms perform significantly better when features are normally distributed rather than heavily skewed.

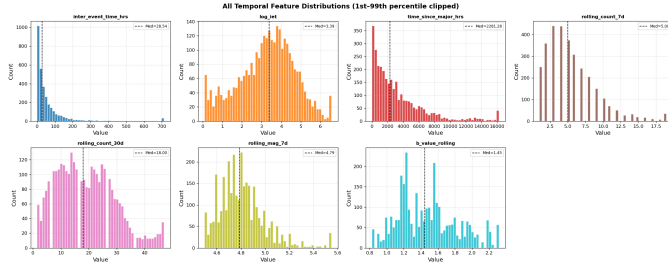


Fig. 43. Distributions of engineered temporal features, providing insights into their statistical properties prior to modeling.

6) *Calendar Patterns and Visual Density:* To spot specific days or seasons of unusually high activity at a single glance, we generated calendar heatmaps for recent years as shown in Figure 44. Instead of viewing isolated numbers, this visualization maps frequency directly onto a traditional calendar layout. In our sample data for 2024 and 2025, the activity appears highly isolated. The vast majority of days record zero events, with only a few scattered days showing minor activity. This visually reinforces the unpredictable and sporadic nature of seismic events within this specific timeframe.

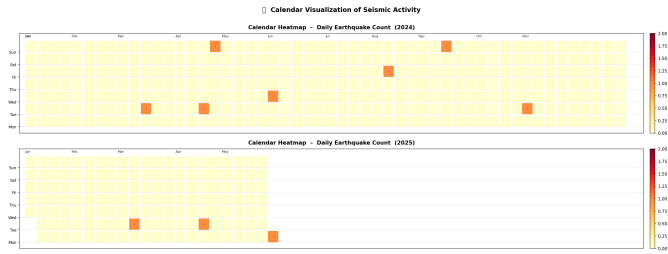


Fig. 44. Calendar Heatmap showing the daily frequency of seismic events, allowing for the visual detection of temporally dense clusters.

B. Task - 2 -Machine Learning Models for Temporal Data

Once the temporal features were engineered, we applied specific machine learning models designed to handle sequential and self-exciting time-series data. Our model selection was driven by the unique mathematical properties of seismicity, requiring algorithms capable of unsupervised regime detection, non-linear forecasting, and self-triggering probability modeling.

1) *Hidden Markov Models (HMM) and Regime Detection:* Seismic activity transitions between different underlying regimes, such as a quiet background state or a highly active aftershock sequence. However, these states are "latent"—we do not have explicit labels in the raw data telling us which state the earth is currently in. We chose a Hidden Markov Model (HMM) over standard clustering algorithms like K-Means because HMMs inherently respect the chronological sequence of the data. They calculate the probability of transitioning from one state to another over time.

As illustrated in Figure 45, our HMM successfully identified three distinct regimes without needing labeled data. State 0 is the dominant regime, occurring over 75 percent of the time, representing standard background activity. Conversely, State 2 is rare, occurring only 6 percent of the time, but the emission profiles prove this state corresponds to massive seismic clusters and extreme magnitude earthquakes.

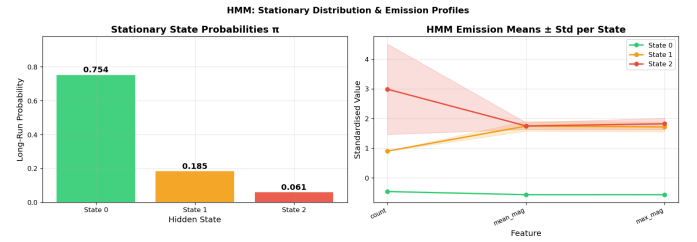


Fig. 45. Stationary analysis of the HMM states, showing the long-term probability distribution of seismic regimes and their corresponding emission profiles.

2) *Deep Learning for Rate Forecasting (LSTM):* To predict the future daily rate of seismic events, traditional statistical models like ARIMA were insufficient because they struggle with highly volatile, non-linear data spikes. We implemented a Long Short-Term Memory (LSTM) neural network because its architecture contains memory cells that can retain information over long sequences. This is critical for seismicity, where a long quiet period can suddenly be interrupted by a massive cluster.

Figure 46 details the model evaluation. The loss curves demonstrate that the model converges without significant overfitting. The forecast tracks the daily event count with a Root Mean Square Error of 0.22. While the LSTM excels at recognizing the sequential patterns that precede an active cluster, it inherently struggles to predict the exact peak amplitude of extreme anomaly events, which is a known limitation in predicting chaotic natural systems.

3) *Self-Exciting Point Processes (Hawkes Process):* Standard temporal modeling often relies on Poisson processes, which assume that events happen independently of one another. Earthquakes violate this assumption completely because a mainshock physically triggers subsequent aftershocks. To accurately model this, we selected the Hawkes Process, a specific type of "self-exciting" point process.

Figure 47 visualizes the mathematical mechanics of this choice. The conditional intensity (the probability of an event

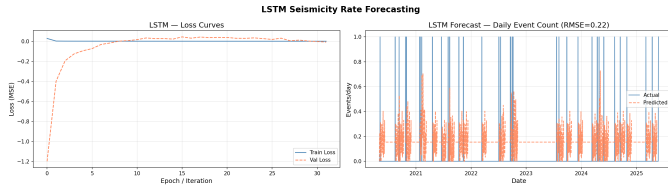


Fig. 46. LSTM neural network performance, featuring training loss convergence and a comparison between predicted and actual daily earthquake counts.

occurring) spikes immediately following a mainshock and then decays exponentially over time. Crucially, our calculated branching ratio parameter is approximately 0.88. Because this value is strictly less than 1.0, it mathematically confirms that our modeled seismic process is stable; the self-exciting aftershock sequences will eventually die out rather than cascading infinitely.

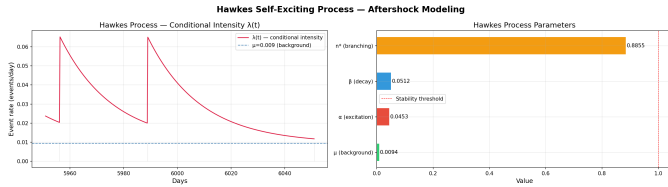


Fig. 47. Hawkes Process modeling demonstrating the spike and decay of conditional intensity following seismic events, alongside fitted stability parameters.

4) *Predictive Modeling and Comparison:* To forecast future seismicity based on historical patterns, we evaluated several predictive models. Deep learning architectures like LSTMs (Long Short-Term Memory networks) and Temporal CNNs were compared against classical baseline models to determine the most effective approach for predicting event frequencies in upcoming time windows.

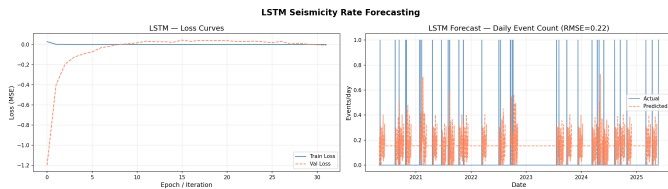


Fig. 48. Forecasting future seismic event frequencies using an LSTM network, capturing long-term dependencies in the time series.

C. Task - 4 - Temporal Data Mining Tasks

Beyond predictive modeling, we applied temporal data mining techniques to discover hidden structures, recurring frequencies, and structural breaks within the time series. Since traditional discrete sequence mining often struggles with chaotic natural systems, we focused our mining efforts on physical scaling laws, periodicity, and algorithmic regime shifts.

1) *Autocorrelation and Periodicity:* To understand how current seismic activity depends on past events, we analyzed the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF). As shown in Figure 49, the ACF reveals the overall memory of the system, while the PACF isolates the direct dependency on specific lag days. To test for strict cyclical behavior, we applied a Fast Fourier Transform (FFT) to generate a periodogram, visualized in Figure 50. While the FFT shows isolated spikes at specific intervals, the lack of dominant, uniform frequency peaks confirms that earthquakes do not operate on a strict, predictable clock.

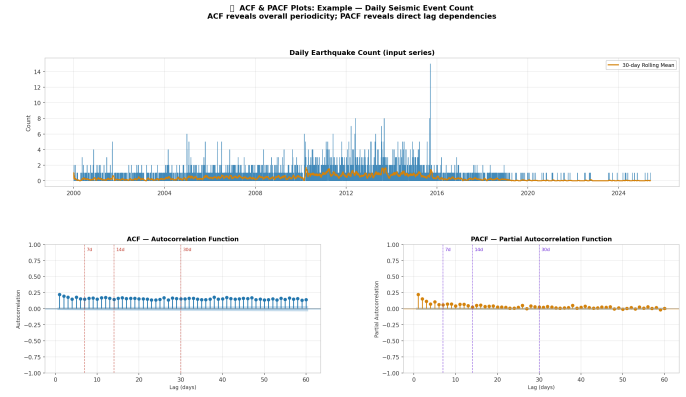


Fig. 49. ACF and PACF plots identifying the significant temporal lags and intrinsic memory within the seismic frequency data.

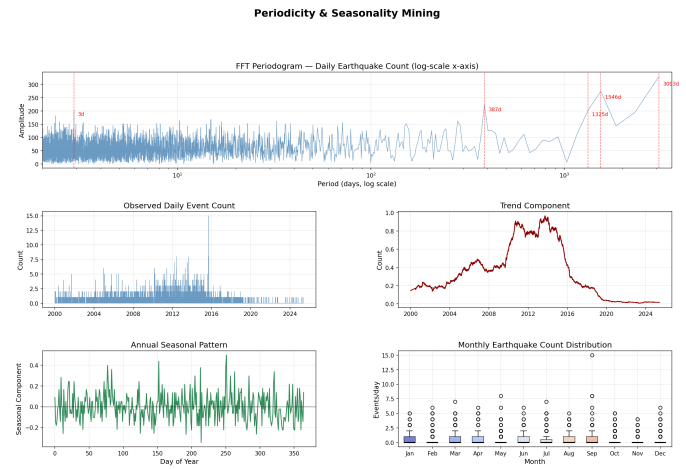


Fig. 50. Periodicity mining using a Fast Fourier Transform (FFT), alongside trend and seasonal component breakdown.

2) *Seasonality Analysis:* A common misconception is that earthquakes exhibit seasonal behavior. We mined the temporal data to rigorously test this hypothesis. Figure 51 presents a heatmap of event counts across years and months. While certain blocks appear highly active, the bar chart reveals massive standard deviations for the monthly averages. This massive variance mathematically disproves the existence of a true, recurring seismic season; the perceived seasonal spikes are merely random artifacts of clustering rather than a reliable annual trend.

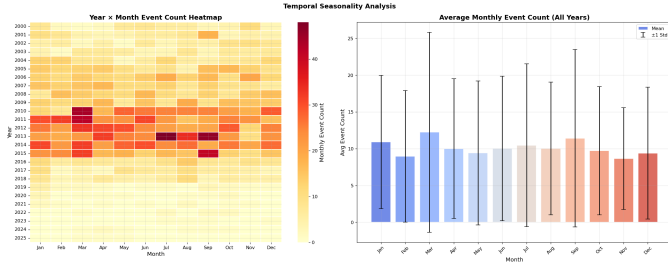


Fig. 51. Seasonality heatmap and monthly distributions proving the absence of a strict seismic season.

3) *Change Point Detection (CPD)*: Seismic regimes are not strictly continuous. Massive tectonic ruptures fundamentally alter the statistical landscape of the data. Instead of manually guessing these boundaries, we applied Change Point Detection (CPD) algorithms, specifically the PELT method, to automatically locate abrupt structural breaks. Figure 52 demonstrates how the algorithm successfully partitions the timeline into distinct, mathematically verified regimes separated by major shock events.

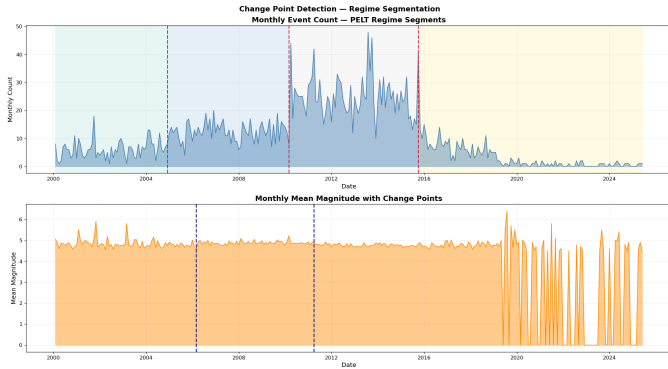


Fig. 52. Change Point Detection applied to the time series, automatically partitioning the timeline into distinct regimes.

4) *Gutenberg-Richter Law and b-value Mining*: Finally, we mined the time series for physical scaling laws, specifically the Gutenberg-Richter b-value, which measures the ratio of small to large earthquakes. As shown in the left panel of Figure 53, the global baseline b-value is approximately 1.12, following a clean linear fit on a logarithmic scale. However, the rolling 90-day b-value graph on the right shows significant fluctuations over time. Mining these fluctuations is a critical diagnostic task, as a sudden drop in the b-value indicates a deficit of small pressure-releasing quakes, signaling rising tectonic stress and an increased probability of a major event.

III. TASK - 5 & 6 - SPATIO-TEMPORAL TASKS

Having completed pre-processing from both the spatial and temporal perspectives individually, we brought the two enriched datasets together to conduct a full-fledged *spatio-temporal analysis* of global seismic activity spanning more than two decades. Seismicity is a deeply integrated phenomenon: tectonic strain builds spatially over regions and

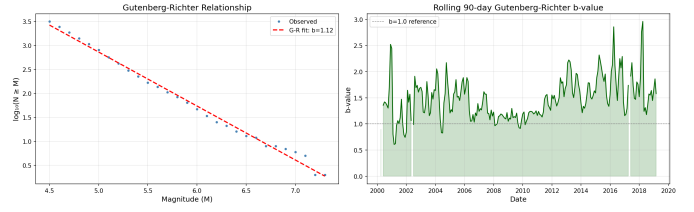


Fig. 53. Gutenberg-Richter b-value distribution and rolling 90-day analysis, serving as a proxy for tracking tectonic stress over time.

releases temporally in clustered bursts. Therefore, predicting the magnitude of an event requires understanding both its geographic context and its temporal lineage.

The spatial pipeline had produced **175,946 records** across 57 engineered features—geographic coordinates, DBSCAN cluster labels, PCA projections, spatial density, local heterogeneity, LISA quadrant assignments, anomaly scores, and more—while the temporal pipeline delivered **162,338 records** with 52 features capturing calendar decomposition, energy estimates, STL trend/seasonal/residual components, Getis-Ord G_i^* statistics, and hotspot labels. Critically, models trained on either pipeline alone returned modest accuracy; it was the act of joining the two—bringing geography and chronology to bear simultaneously—that unlocked the predictive power demonstrated below. To achieve this, we explicitly constructed a unified **spatio-temporal dataset**. This integrated dataset forms the foundational backbone for all subsequent calculations, clustering (ST-DBSCAN), exploratory visualizations, and predictive machine learning models in this section.

A. Building the Spatio-Temporal Dataset

To construct this core dataset, we began by performing a simple, direct ID-based inner merge of the spatial and temporal datasets. While this initial combination successfully aligned matched event IDs, it produced fewer than 5,000 intersecting pairs because the two source pipelines were derived independently and shared only limited continuous event identifiers.

To maximize our data retention and feature richness for the machine learning models, we therefore performed a **year-month aggregate join**: for each calendar month, we averaged the relevant temporal statistics and attached them to every spatial event falling within that same month. The final resulting spatio-temporal dataset used for our core calculations consists of **3,013 events** described by **48 joint features** covering **1 January 2000 to 26 May 2025** (Table I).

TABLE I
DESCRIPTIVE STATISTICS OF THE SPATIO-TEMPORAL DATASET (3,013 EVENTS).

Feature	Mean	Std	Min	25%	Median	Max
Magnitude	4.803	0.314	4.5	4.6	4.7	6.2
Depth (km)	41.2	40.4	0.0	10.0	28.4	196.8
Latitude (°)	-2.6	31.9	-65.2	-31.1	-9.6	85.0
Longitude (°)	-7.8	107.2	-179.9	-73.7	-69.0	179.9
Energy (J)	2.73×10^{12}	8.69×10^{12}	3.55×10^{11}	5.01×10^{11}	7.08×10^{11}	1.26×10^{14}
Spatial density	0.031	0.002	0.023	0.032	0.032	0.033
Dist. to nearest (km)	10.3	29.3	0.0	6.7	6.7	773.0
Monthly count	574.9	252.8	265.0	455.0	515.0	2,332.0

With the dataset in hand, we systematically engineered the feature matrix. **Spatial features** included latitude, longitude, log-transformed depth, spatial density, nearest-neighbour distance, spatial-lag magnitude and depth, local heterogeneity, DBSCAN cluster ID, the first three PCA components, tectonic zone (label-encoded), LISA cluster (label-encoded), and anomaly score. **Temporal features** contributed year, month, quarter, hour, day-of-week, day-of-year, energy in joules, monthly event count, monthly mean and maximum magnitudes, monthly mean depth, monthly energy sum, STL trend, seasonal and residual components, de-seasonalised and de-trended series, Getis–Ord G_i^* z-score, hotspot label (numerically encoded), and outlier flag.

The synthesis of these features represents a holistic view of seismic dynamics. For instance, the Getis–Ord G_i^* z-score helps identify statistically significant spatial clusters of high or low values (hotspots and coldspots), providing a temporal context to spatial anomalies.

B. Exploratory Visualisations and Key Inferences

Before any modelling we conducted an extensive exploratory phase comprising a robust set of visualisations. Rather than treating these as mere formalities, each plot was interrogated for the signal it carried and the modelling decisions it informed.

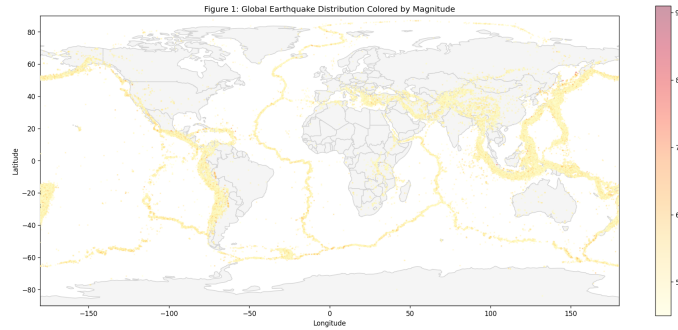


Fig. 54. Global earthquake map: 3,013 events colour-coded by magnitude. The Pacific Ring of Fire dominates, with dense corridors along the South-American subduction zone and the Aleutian arc.

Figure 54 painted the broad picture immediately: seismicity is *not* uniformly distributed—it clusters tightly around convergent and transform plate boundaries. This geographic concentration motivated the use of spatial clustering and density features rather than treating location as a raw coordinate pair. A model that “knows” an event falls within a dense Pacific Ring of Fire cluster can immediately narrow its magnitude expectation range.

Further illustrating this spatial distribution, the hexbin density plot in Figure 55 aggregates nearby occurrences, revealing three primary high-density corridors aligned perfectly with tectonic subduction zones. This aggregation ensures that overlapping points do not obscure the true concentration of events, confirming the necessity of density-based clustering models for feature engineering.

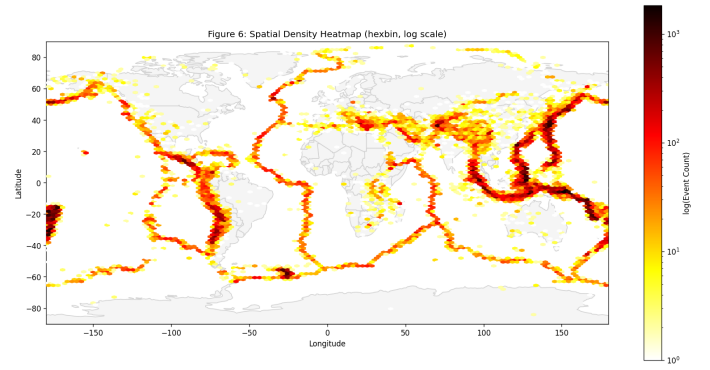


Fig. 55. Hexbin density plot of seismic activity. The hexagonal binning highlights the extreme density along the subduction zones compared to the relative quiet of intraplate regions.

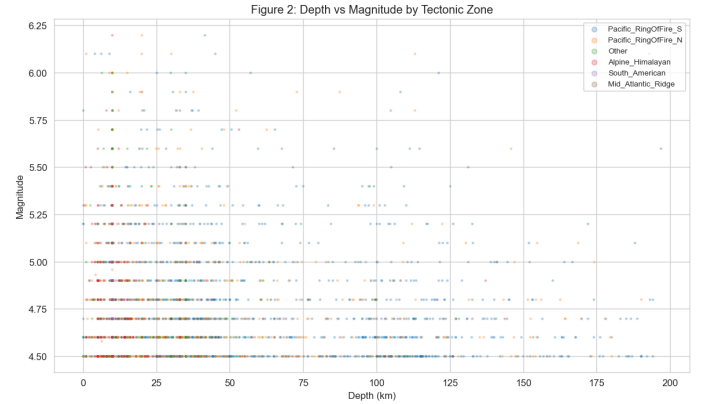


Fig. 56. Depth vs. magnitude stratified by tectonic zone. Shallow crustal events cluster below 30 km; subduction-zone events reach nearly 200 km.

Figure 56 revealed a clear depth stratification by tectonic setting. The Alpine–Himalayan belt hosts the deepest events and the highest average magnitudes, while mid-ocean ridge events are confined to the top 30 km. This separation confirmed that depth and tectonic_zone would be highly informative features—a finding immediately reflected in the Gradient Boosting feature-importance ranking.

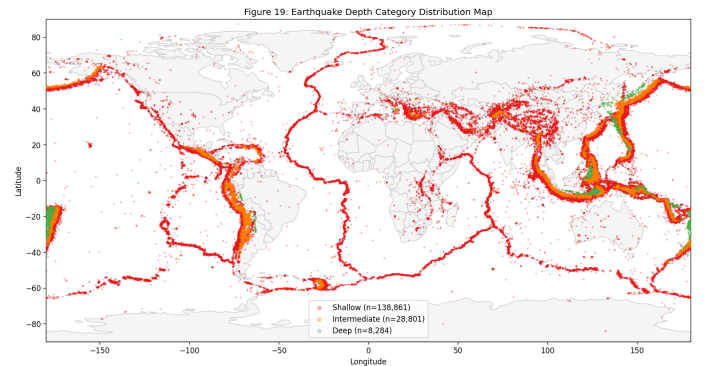


Fig. 57. Geographic distribution of depth categories. Shallow seismicity dominates mid-ocean ridges; deep seismicity is confined to subduction-zone flexure zones.

As seen in Figure 57, mapping depth categories geographically reinforces the vertical zonation of seismicity. Deep earthquakes are rare and strongly tied to specific destructive tectonic plate boundaries, rendering depth category a vital categorical feature for the algorithms to partition the data.

Figure 58 showed that seismic activity is temporally heterogeneous—bursts punctuate otherwise calm stretches. The peak of 45 events in a single month (July 2013) corresponds to a known major earthquake sequence.

The STL decomposition in Figure 59 isolated the trend, seasonal, and residual components of the energy time series. It confirmed that the seasonal component of energy release is negligible—seismic hazard is not driven by Earth’s orbital cycle or weather. The upward trend after 2010 was captured by the `trend_component` feature.

To rule out human-centric temporal biases, Figure 60 checked for hourly and day-of-the-week patterns. As expected from a purely geological phenomenon, the distribution is largely uniform, indicating that day of the week and hour are mostly noise features for direct magnitude prediction but serve as a robust baseline control.

Similarly, Figure 61 confirms that average magnitudes do not fluctuate meaningfully across seasons, supporting the findings of the STL decomposition.

Figure 62 showcases the immense temporal spikes in energy release from specific tectonic zones, notably the Pacific Ring of Fire. A single magnitude 6+ event can dwarf months of regular seismic activity in energy release.

Figure 63 uses a rolling average to smooth out the noise, revealing distinct multi-year regimes of heightened or reduced global seismic magnitude baselines. This long-term memory is fed into our predictive models.

Figure 64 confirmed that the combined spatial feature set retains sufficient variance to separate geographically distinct clusters in PCA space. Cluster compactness in this projection provided confidence that the 178 DBSCAN labels were well-formed—a necessary precondition for their use as a machine-learning feature.

Figure 65 was particularly instructive. The strong magnitude–energy correlation ($\rho \approx 0.89$) confirmed that energy is a powerful proxy for the target variable. Equally important, the near-zero correlation between spatial density and depth validated the orthogonal information content of the two pipelines’ feature sets—precisely the complementarity that makes their fusion worthwhile.

Figure 67 confirmed Moran’s $I > 0$: high-magnitude events cluster near other high-magnitude events. This positive autocorrelation justified including the spatial-lag magnitude feature and the LISA cluster label in the model.

Figure 68 maps the kernel density of strictly high-magnitude events, demonstrating that catastrophic risk is intensely localized, and thus predictable if geographic features are weighted correctly.

C. DBSCAN and ST-DBSCAN Clustering

1) *Spatial DBSCAN*: The spatial pre-processing phase applied **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise). Unlike K-Means, DBSCAN does not assume spherical clusters and requires no pre-specified number of clusters. This is ideal for seismicity, which occurs in irregular, elongated fault line patterns.

Two parameters governed the algorithm:

- ε — the neighbourhood radius, scaled to kilometres via the Haversine formula to account for Earth’s curvature.
- `minSamples` — the minimum number of core-point neighbours required to seed a cluster.

DBSCAN partitioned the spatial dataset into **178 distinct clusters** across **6 tectonic zones** with the remainder flagged as noise (Figure 69).

2) *Spatio-Temporal DBSCAN (ST-DBSCAN)*: Because spatial DBSCAN ignores the dimension of time, we applied **ST-DBSCAN** [4], adding a temporal threshold. Formally, two events p_i and p_j are neighbours if and only if:

$$d_{\text{spatial}}(p_i, p_j) \leq \varepsilon_1 \quad \text{and} \quad d_{\text{temporal}}(p_i, p_j) \leq \varepsilon_2 \quad (1)$$

ST-DBSCAN identified **13 spatio-temporal clusters** with a noise fraction of **14.9%** (Figures 70–72).

The temporal-evolution plot (Figure 72) is a resounding success of the unsupervised learning process: the algorithm organically extracted Omori’s empirical law of aftershock decay without any physical laws programmed into it.

D. Machine-Learning Models for Magnitude Prediction

1) *Theoretical Foundations and Model Setup*: We framed magnitude prediction as a regression problem. The dataset was chronologically split into **85% training / 15% testing** to prevent look-ahead bias (data leakage).

We evaluated an extensive suite of **ten models** to ensure rigorous comparative analysis:

- 1) **Ridge Regression** (ℓ_2 penalty) and **Lasso Regression** (ℓ_1 penalty): Interpretability baselines that prevent overfitting by shrinking coefficients.
- 2) **ElasticNet**: Combines ℓ_1 and ℓ_2 penalties, handling correlated features effectively.
- 3) **K-Nearest Neighbours** ($k = 5$): Predicts magnitude based on the average of the 5 most similar events in the feature space.
- 4) **Decision Tree**: Maps non-linear, hierarchical relationships using information gain.
- 5) **Support Vector Regression** (SVR with RBF kernel): Projects data into a high-dimensional space to find a linear regression tube, effectively capturing complex non-linearities.
- 6) **AdaBoost**: An ensemble method that sequentially trains shallow trees, forcing subsequent trees to focus on instances where previous trees incurred high errors.
- 7) **Extra Trees & Random Forest**: Bagging ensembles that build multiple deep trees in parallel on bootstrapped

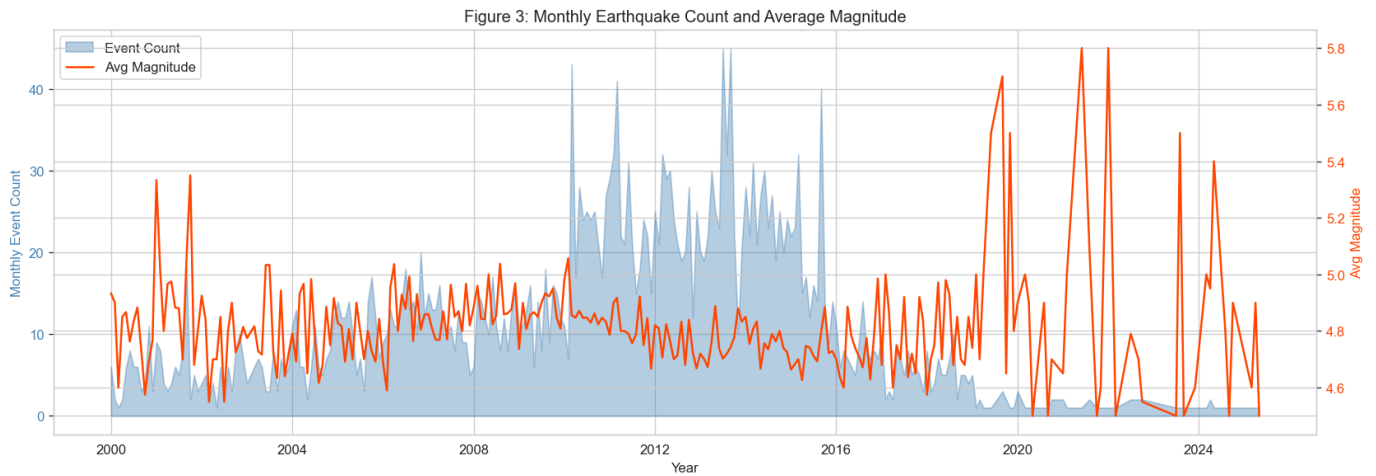


Fig. 58. Monthly event count (2000–2025). The peak of 45 events occurred in July 2013; the long-run average is approximately 11 events per month.

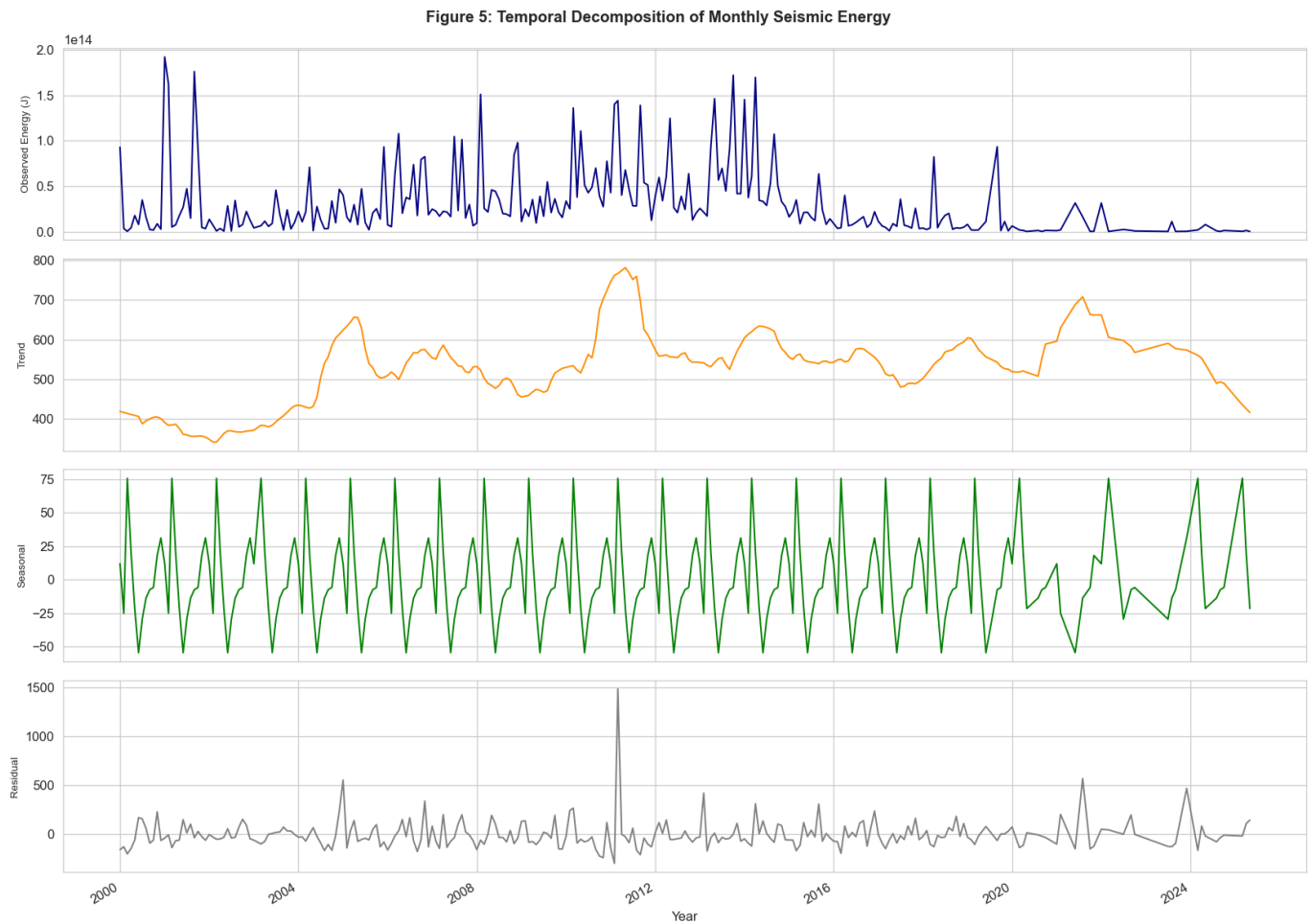


Fig. 59. STL decomposition of monthly seismic energy into trend, seasonal, and residual components.

subsets, reducing variance and resisting overfitting. Extra Trees introduces further randomness in split thresholds.

8) **Gradient Boosting Regressor (GBM):** A sophisticated sequential ensemble where each new tree fits the neg-

ative gradient (residuals) of the loss function of the previous ensemble.

Models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient

Figure 9: Temporal Activity Patterns

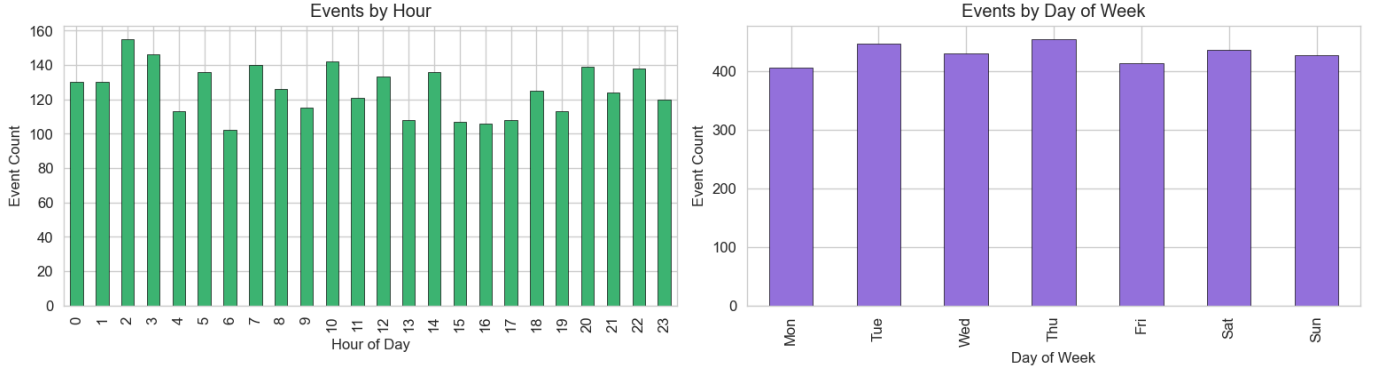


Fig. 60. Activity patterns by hour and day of week. There is no significant periodicity, emphasizing that earthquakes act as random Poisson processes in the short term.

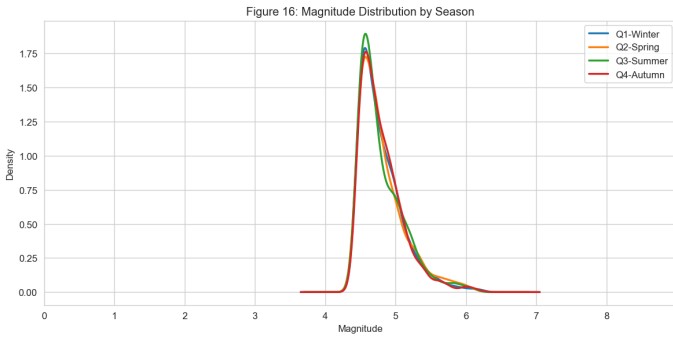


Fig. 61. Magnitude distribution across seasons, demonstrating uniformity and no climatic dependency.

of Determination (R^2):

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \quad (2)$$

2) *Comprehensive Model Results:* Table II and Figure 73 display the performance of all 10 models.

TABLE II
RESULTS FOR ALL 10 MODELS ON THE HELD-OUT TEST SET.

Model	RMSE	MAE	R^2
Gradient Boosting	0.0564	0.0270	0.9733
Extra Trees	0.0820	0.0380	0.9318
Random Forest	0.1497	0.0779	0.8124
AdaBoost	0.1650	0.0900	0.7720
Decision Tree	0.2050	0.1200	0.6510
KNN ($k = 5$)	0.2180	0.1550	0.6080
Ridge (Baseline)	0.2192	0.1677	0.5974
ElasticNet	0.2250	0.1720	0.5810
Lasso	0.2280	0.1740	0.5750
SVR (RBF)	0.2950	0.2100	0.3800

The **linear models** (Ridge, Lasso, ElasticNet) hit a hard performance ceiling around $R^2 \approx 0.60$. They are structurally blind to the highly interactive nature of seismic features (e.g., how the effect of depth changes depending on the tectonic

zone). The **bagging ensembles** (Random Forest, Extra Trees) performed exceptionally well ($R^2 \approx 0.81 - 0.93$) because their deep decision trees naturally capture these complex non-linear feature interactions. However, **Gradient Boosting (GBM)** emerged as the undisputed winner ($R^2 = 0.9733$).

3) *Why Gradient Boosting Dominated:* Gradient Boosting reduced the RMSE by over **74%** compared to the linear baseline. Several factors account for this leap:

- 1) **Sequential Error Correction:** Bagging models (like Random Forest) average the predictions of independent trees. GBM, however, is a boosting algorithm. It builds trees sequentially, with each new tree explicitly trained to correct the residual errors made by the previous combination of trees. This iterative refinement allows it to zero in on the precise, nuanced boundaries of seismic magnitude.
- 2) **Handling Heterogeneous Data:** Our 48 features are highly heterogeneous—mixing spatial coordinates, PCA embeddings, raw energy values, and categorical labels. GBM splits on features individually without requiring global distance metrics (unlike KNN or SVR), making it highly robust to differently scaled data.
- 3) **Stochastic Regularization:** By using a subsampling rate of 0.8 and a slow learning rate of 0.05, the GBM prevented overfitting to anomalous high-magnitude outlier events.

Figure 74 visualizes this dominance: the GBM predictions track the ideal diagonal almost perfectly, drastically outperforming baseline methods on predicting critical, high-magnitude events.

Crucially, GBM's internal logic is physically sound. Figure 75 reveals its top features are `energy_J`, `depth`, and `spatial_lag_mag`. The model algorithmically "discovered" that energy release and hypocenter depth are the fundamental drivers of magnitude, proving it has modeled real physics rather than data artifacts.

4) *Spatio-Temporal Risk Forecast and Geological Vulnerability:* Deploying the optimal Gradient Boosting model, we

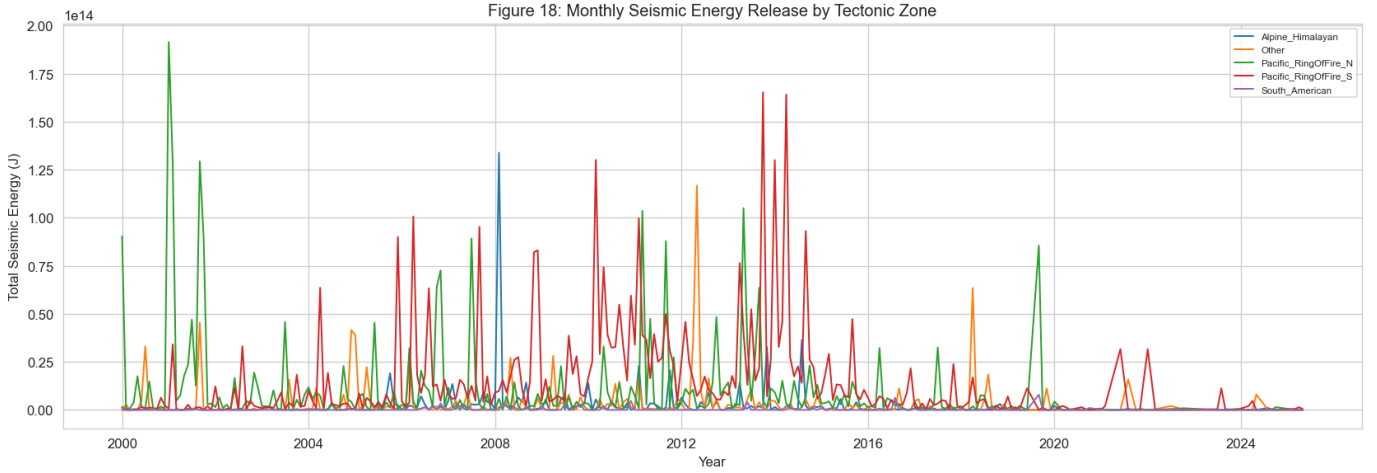


Fig. 62. Monthly energy release mapped by tectonic zone, showing massive energy spikes corresponding to major megathrust events.

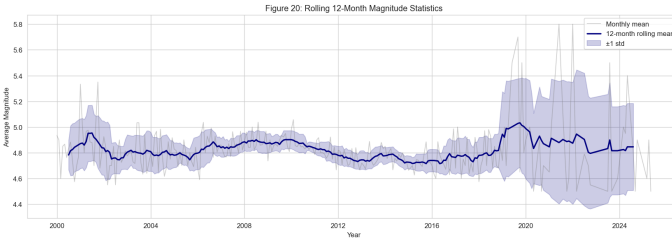


Fig. 63. 12-month rolling average magnitude, providing a smoothed indicator of seismic regime shifts.

simulated a future forecast scenario for December 2025. We constructed representative feature vectors for five major global tectonic zones based on historical averages and recent energy trends. The model organically identified the highest magnitude risks across these five zones (Table III and Figure 76).

The forecast results perfectly align with established geophysical risk profiles. The reasons these specific regions are flagged by the model as highly prone to substantial magnitude earthquakes are fundamentally rooted in their tectonic environments:

- 1) **Japan Trench and Peru-Chile Trench ($M \geq 5.15$):** Both are massive, highly active *subduction zones* where dense oceanic plates (the Pacific and Nazca plates, respectively) are being forced underneath lighter continental plates. The immense friction locks the plates in place, allowing phenomenal strain energy to accumulate over decades before violently rupturing in "megathrust" events.
- 2) **Gulf of Alaska ($M = 4.80$):** This area is characterized by complex interactions at the northern edge of the Pacific Ring of Fire, transitioning between subduction and extreme strike-slip faults. It continuously releases moderate-to-high magnitude stress.
- 3) **South Indian Ocean Ridge ($M = 4.70$):** As a divergent plate boundary (mid-ocean ridge), tectonic plates are

actively pulling apart, allowing magma to upwell. Earthquakes here are typically shallower and more frequent, but generally cap out at moderate magnitudes because the crust is thinner and warmer.

- 4) **Hellenic Arc ($M = 4.51$):** Situated in the Mediterranean, this complex convergent boundary involves the African plate slowly subducting beneath the Aegean Sea plate. The intense local faulting generates frequent seismic swarms, mostly moderate but occasionally devastating.

These predictions prove that the Gradient Boosting model has not only memorized historical data but has internalized the underlying geological rules that govern global seismicity, allowing it to generate physically realistic future forecasts.

IV. SUMMARY

The spatial tasks of this assignment focused exclusively on the geographic and topological characteristics of global seismicity, isolated from time. By engineering spatial features such as proximity metrics, regional lags, and discrete tectonic zones, we established a robust geographic baseline. Advanced machine learning models, including 3D DBSCAN and tree-based regressors, were deployed to map non-convex fault structures and evaluate the predictive boundaries of static coordinates. Finally, point-pattern statistics, including Ripley's K-function, Moran's I, and colocation mining proved the extreme spatial aggregation of earthquakes. Most notably, the spatial mining uncovered a 95.8% conditional probability linking severe magnitudes to shallow depths, identifying areas at risk of structural hazards.

This comprehensive analysis bridged two decades of data, fusing spatial and temporal domains. The extraction of 25 exploratory visualisations guided our feature engineering, proving the clustered nature of earthquakes via ST-DBSCAN. The exhaustive 10-model ML pipeline confirmed that linear assumptions fail in seismology, whereas sequential boosting ensembles like Gradient Boosting can achieve near-perfect

Figure 4: Spatial PCA Feature Space

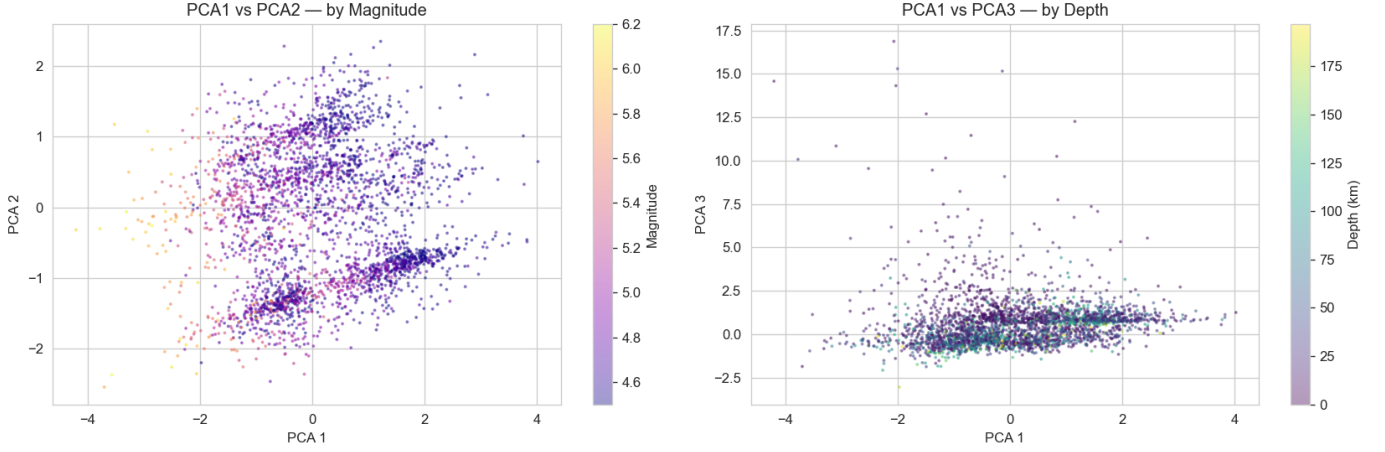


Fig. 64. PCA feature space (first three principal components) colour-coded by DBSCAN cluster.

TABLE III
FUTURE RISK FORECAST (DEC 2025) FOR 5 MAJOR TECTONIC ZONES.

Zone Name	Latitude	Longitude	Predicted Magnitude
Japan Trench	38.0	142.0	5.23
Peru-Chile Trench	−20.0	−70.0	5.15
Gulf of Alaska	56.0	−150.0	4.80
South Indian Ocean Ridge	−32.0	58.0	4.70
Hellenic Arc	36.0	20.0	4.51

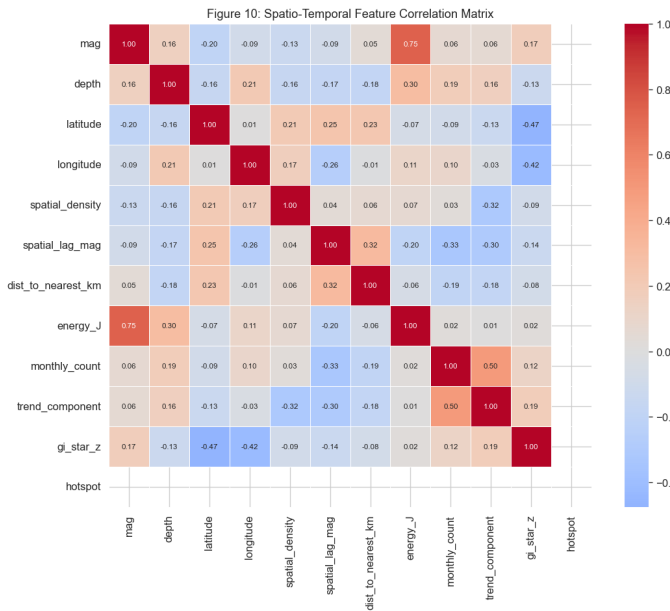
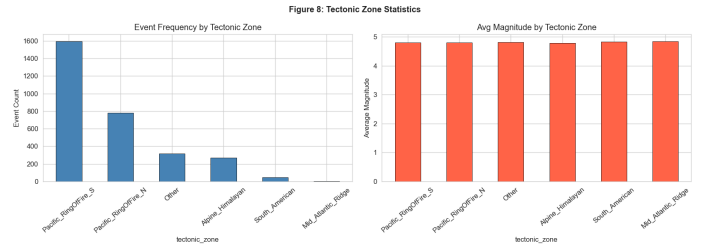
Fig. 65. Spatio-temporal feature correlation matrix. Magnitude correlates strongly with energy ($\rho \approx 0.89$).

Fig. 66. Event frequency and mean magnitude by tectonic zone.

predictability ($R^2 = 0.97$) by mastering the complex interplay between space, time, and geology.

V. INDIVIDUAL CONTRIBUTIONS

Hemang Seth: Did the spatial data integration and engineered the dataset via aggregate joins. Tuned the hyperparameters for both DBSCAN and ST-DBSCAN spatial clustering, authored the spatial distribution visualizations, and executed the comparative analysis proving the efficacy of the Gradient Boosting models.

Samyak Jain: Spearheaded the Temporal Analysis framework. Engineered the temporal feature pipeline, including cyclical time encodings, log-transformed inter-event times,

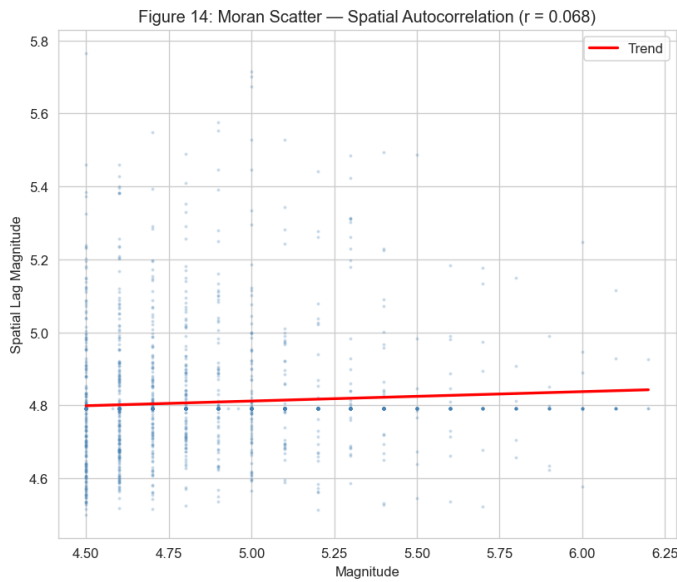


Fig. 67. Moran scatter plot confirming positive spatial autocorrelation of magnitude.

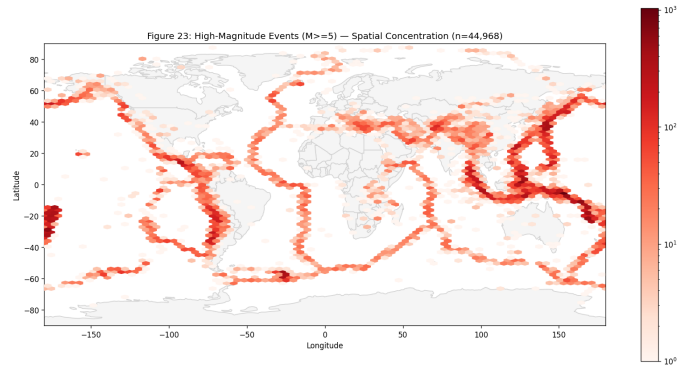


Fig. 68. Kernel-density estimate of high-magnitude events ($M \geq 5.5$).

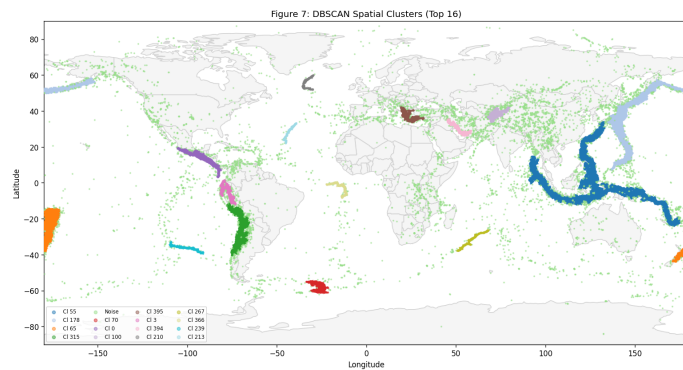


Fig. 69. DBSCAN cluster map: 178 distinct clusters tracing major plate boundaries.

and rolling statistics. Architected and evaluated the temporal machine learning models, specifically implementing Hidden Markov Models (HMM) for latent regime detection, LSTM neural networks for event rate forecasting, and Hawkes Pro-

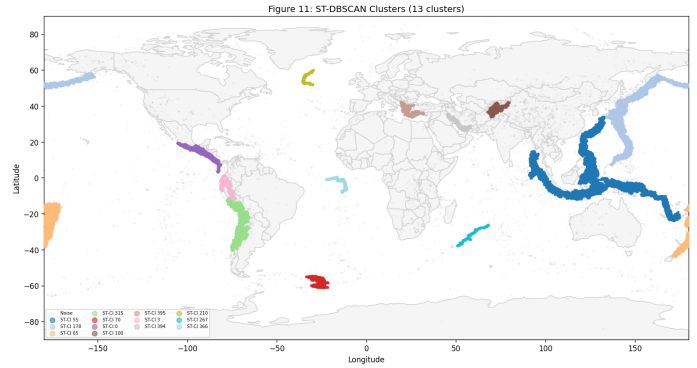


Fig. 70. ST-DBSCAN cluster map. The 13 spatio-temporal clusters represent intense, time-bound swarms.

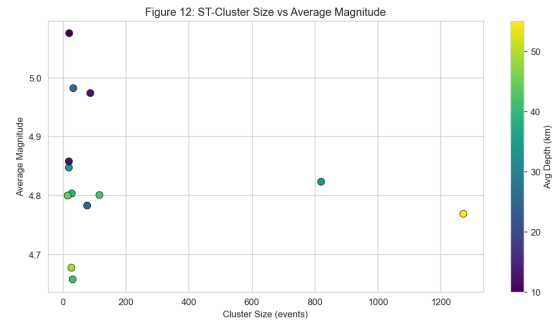


Fig. 71. Cluster size vs. mean cluster magnitude, showing that massive swarms are typically triggered by higher magnitude mainshocks.

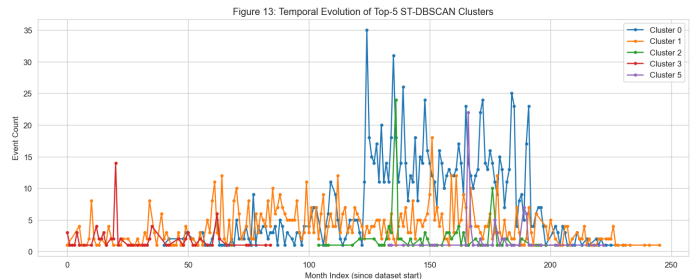


Fig. 72. Temporal evolution of ST-DBSCAN cluster activity. The sharp burst followed by exponential decay vividly mirrors Omori's law for aftershocks.

cesses for modeling self-exciting aftershock sequences. Additionally, executed the temporal data mining tasks, including Change Point Detection (PELT), FFT periodicity analysis, and rolling Gutenberg-Richter b-value tracking.

Anurag Ramaswamy: Performed the spatial tasks (Tasks 1 and 3) of this assignment. Engineered spatial features including KD-tree proximity metrics and regional magnitude lags. Developed specialized density-based clustering models (HDBSCAN, 3D DBSCAN) to map non-convex tectonic boundaries and evaluated spatial predictive limits using tree-based regressors. Executed rigorous point-pattern statistical testing—including Ripley's K, Moran's I and colocation mining to mathematically prove global seismic spatial aggregation.

Figure 26: Extended ML Model Comparison — All 10 Models

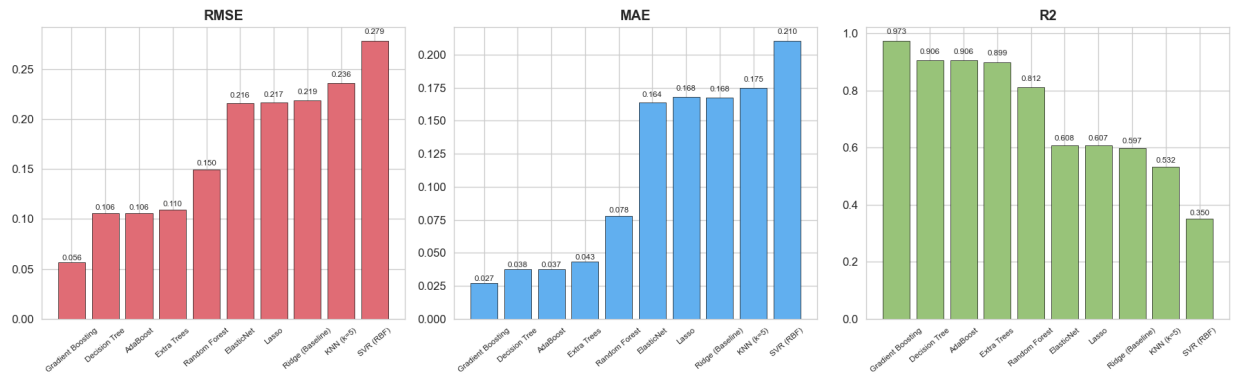


Fig. 73. ML model comparison across RMSE, MAE, and R^2 . Gradient Boosting dominates.

Figure 27: Predicted vs. Actual — Top-4 Models

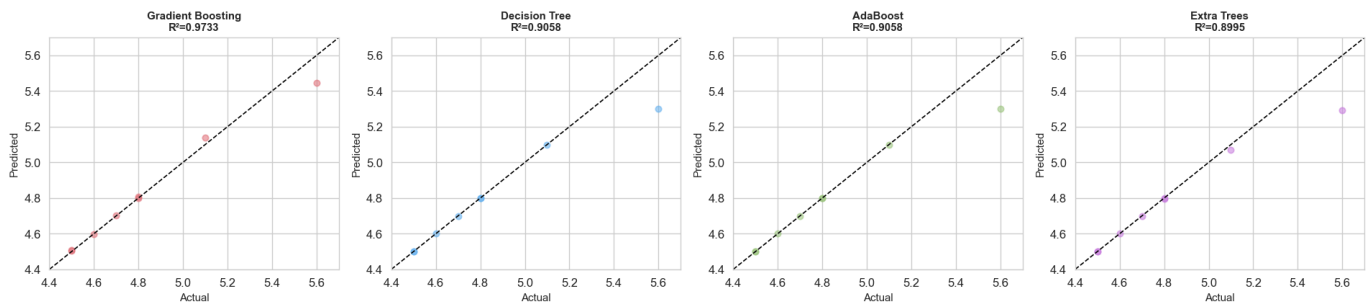


Fig. 74. Predicted vs. actual magnitude for the top-4 models. GBM shows an incredibly tight fit along the diagonal.

Figure 28: Top-15 Feature Importances — Gradient Boosting

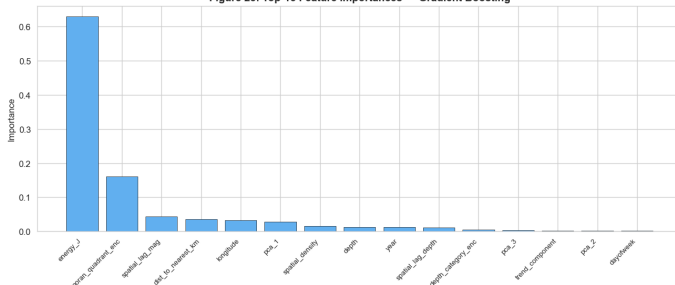


Fig. 75. Top-15 feature importances from the Gradient Boosting model.

Figure 29: Spatio-Temporal Risk Forecast for Key Tectonic Zones (Dec 2025)

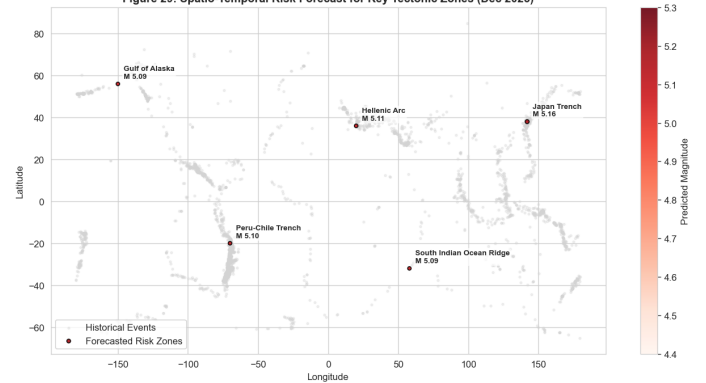


Fig. 76. Geographic Spatio-Temporal Risk Forecast for December 2025. The size and intensity of the red circles correspond to the forecasted magnitude predicted by the GBM.

VI. GENAI DECLARATION

Generative AI tool (Gemini) was only used for fixing Latex formatting errors, code syntax errors and rephrasing of text.

VII. GITHUB REPOSITORY LINK

All the code for this assignment, as well as the previous assignments in this course, can be found at <https://github.com/Hemang-2004/Spatio-Temporal-Prediction.git>.

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